Integrable semi-discretization and numerical approximation of the (2+1)-dimensional dispersive long-wave system

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Abstract

Exact solutions of the $(2+1)$-dimensional dispersive long-wave (DLW) system provide a detailed description of nonlinear wave propagation and interaction. Extending these solutions to a spatial lattice brings their integrable structure into a discrete setting and raises the question of how the lattice changes the physical waves. In this work, we develop two related integrable semi-discretizations to study this connection between lattice construction and wave approximation. Starting from the Hirota–Miwa equation, a Miwa expansion gives an endpoint formulation (Endpoint-SD). A half-step parameter shift and a staggered placement of the τ functions then yield a centered formulation (Centered-SD). Both admit explicit Gram determinant solutions and nonlinear physical-field equations with compatible Darboux–Lax representations, allowing the effects of lattice placement to be studied within a common integrable framework. To quantify these effects, we first examine how the equations and their exact wave fields approach their continuous counterparts. Endpoint-SD is second-order consistent in the first field equation and first-order consistent in the second, while Centered-SD achieves second-order consistency in both. Under suitable regularity and positivity assumptions, we further establish second-order approximation of the exact Centered-SD wave fields. Together with the leading correction to the Endpoint-SD equation residual, this relates the lattice construction to its continuum approximation. These analytical comparisons provide a basis for numerical experiments with soliton, Gaussian and sinusoidal profiles. Through these experiments, we show how lattice placement, wave profile and field reconstruction jointly influence computational accuracy, with the consistency analysis helping to interpret the behavior at finite lattice spacing. Building on this spatial comparison, we then examine how the errors develop during numerical evolution under different time integrators. The semi-discrete structure also offers a way to improve the resolution of the waves. Specifically, we establish a conservation law and use it to construct a self-adaptive moving mesh, improving the accuracy of both fields in the soliton tests at a fixed node count. These results clarify how the arrangement of an integrable lattice and the reconstruction of its physical fields influence the approximation of continuous waves. By tracing these choices from the bilinear equations to the evolving field errors, the study provides a basis for understanding and improving numerical accuracy within a common integrable framework.

Keywords: Dispersive long-wave system; Integrable discretization; Bilinear method; Self-adaptive moving mesh method; Consistency analysis

1 Introduction

The propagation and interaction of shallow-water waves have long attracted interest in the study of nonlinear wave phenomena. Nonlinearity and dispersion shape the evolving wave profiles, and their balance can support coherent waves that persist during propagation. Nonlinear long-wave equations describe these effects in a reduced form, allowing wave motion to be studied without retaining the full complexity of the underlying fluid equations. Within this class, integrable models provide exact solutions through which individual waves and their interactions can be examined, together with bilinear formulations and Lax representations that describe the structures governing their evolution. These analytical results also provide reference solutions for

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numerical computation. Representing the waves on a spatial lattice introduces choices that affect both their approximation and the structure of the governing equations. Integrable discretization brings these questions together by constructing discrete equations alongside their exact solutions and continuum limits.

The $(2+1)$-dimensional dispersive long-wave (DLW) system extends the dispersive long-wave equation to two spatial dimensions [1] and has been studied in connection with wave motion in wide channels and open seas of finite depth [3,4]. Since its introduction by Boiti, Leon and Pempinelli, its solutions have been investigated through several related analytical approaches. The inverse spectral transform, Bäcklund transformation and superposition formula established in the original work [1] were followed by nonlinear transformations [2] and bilinear methods [3], providing further ways to construct wave solutions and understand their relations. Determinant representations give these solution families an explicit form. Hu, Xu and Yu developed Grammian and Wronskian solutions [4], while Sheng and Yu used a KP-hierarchy reduction to obtain Gram determinant solutions describing solitons, line breathers and rational waves on constant and periodic backgrounds [5]. Their bilinear formulation and determinant construction provide the continuous starting point for the present study.

The bilinear relations underlying these solutions also offer a starting point for introducing a spatial lattice [15]. Bäcklund transformations can relate neighbouring lattice values, while expansions of the Hirota–Miwa equation [17,18] connect discrete bilinear relations to equations with continuous variables. These approaches have been developed for semi-discrete soliton equations [7] and extended to systems with two spatial dimensions [8]. For the DLW system, a further connection comes from its bilinear relation to the $(2+1)$-dimensional sinh-Gordon equation [5], whose integrable discretizations admit determinant solutions and Bäcklund–Lax structures [6]. Together with earlier studies of discrete DLW systems [11], these developments provide a basis for carrying continuous wave families onto a spatial lattice. They also invite a closer examination of what this passage means for the physical waves, whose approximation depends on both the lattice equations and the reconstruction of the physical fields.

Building on this work, we wish to understand how the preservation of integrable structure can contribute to the approximation and computation of continuous waves. We seek a semi-discrete setting in which the bilinear equations, exact solutions, and reconstructed physical fields can be studied together, allowing us to trace the effect of introducing a lattice from the analytical construction to the resulting wave profiles. In particular, we ask how different lattice arrangements within a common integrable framework affect the accuracy of the physical fields, and how the available analytical structure can guide their numerical computation. These questions connect the construction of integrable lattices with the broader task of understanding and improving numerical approximations of nonlinear waves.

To investigate these questions, we construct two integrable semi-discretizations of the DLW system from the Hirota–Miwa equation. The endpoint construction, denoted by Endpoint-SD, follows from a Miwa expansion. Starting from this lattice, we introduce a shift of the parameter reference and a staggered placement of the τ functions to obtain the centered construction, denoted by Centered-SD. We establish their Gram determinant solutions and Darboux–Lax representations, and show how the lattice arrangement enters the reconstructed physical-field equations. Endpoint-SD has second-order consistency in the first equation and first-order consistency in the second, whereas Centered-SD achieves second-order consistency in both. We derive the leading correction to the Endpoint-SD equation residual and establish second-order approximation of the exact Centered-SD wave fields under the stated assumptions. These results show how two constructions with shared integrable structures can produce different approximation properties, and identify the role of lattice placement and field reconstruction in determining those properties.

The exact solutions provide benchmarks for examining how the analytical approximation properties are reflected in numerical computation. We compare initial time-derivative errors and follow the physical-field errors through time on soliton, Gaussian, and sinusoidal backgrounds. These experiments, together with the spatial error analysis, help explain how the formulation and time integrator affect the accuracy of the evolving waves. The semi-discrete conservation law also supports a shared-node moving mesh, which reduces the errors in both fields at a fixed node count in the soliton tests. Independent initial perturbations extend the study to the development of coupled responses within each formulation. Through these investigations, we seek to clarify how the analytical structure of an integrable semi-discretization can help us understand numerical behavior and improve the computation of nonlinear waves.

The paper is organized as follows. Section 2 introduces the continuous DLW bilinear system and develops the Endpoint-SD and Centered-SD constructions, together with their Gram determinant solutions and positivity conditions. Section 3 reconstructs the nonlinear physical-field equations, establishes their

Darboux–Lax representations, and examines the continuum limits of the equations and exact wave fields. Building on these formulations, Section 4 describes the spatial discretization, initial and boundary data, time integration, and conservation-based moving mesh. Section 5 presents the numerical experiments on physical-field accuracy, mesh adaptation, and the evolution of independent initial perturbations. Section 6 summarizes the main findings.

2 Integrable bilinear semi-discrete formulations and exact solutions

In this section, we develop a bilinear lattice description of the DLW system that connects the continuous equations, the arrangement of the τ functions and the construction of exact solutions. In Section 2.1, we recall the continuous system and its bilinear representation, fixing the parameters and the transformation to the physical fields. In Section 2.2, we derive an unreduced bilinear lattice from the Hirota–Miwa equation and construct a unified Gram determinant family using a standard identity of the KP hierarchy. Section 2.3 specializes this construction to the endpoint and centered formulations, compares their bilinear consistency, and gives a positive real solution class.

2.1 The continuous DLW system and its bilinear representation

We begin with the continuous DLW system and its bilinear representation. Specifically, we seek a description in which the physical fields and their explicit solutions can be obtained from a common pair of τ functions. With real parameters a and λ, the continuous equations are

$$u_{yt} + v_{xx} + [(u+2a)u_y]_x = 0, \qquad v_t + u_{xxy} + [(u+2a)v + 2\lambda u]_x = 0. \tag{1}$$

For positive, sufficiently smooth functions f, g, we use the transformation

$$u = 2\partial_x \log(f/g), \qquad v = 2\partial_{xy} \log(fg). \tag{2}$$

Hirota's bilinear derivatives are defined by

$$D_x^m D_y^n D_t^\ell f \cdot g = (\partial_x - \partial_{x'})^m (\partial_y - \partial_{y'})^n (\partial_t - \partial_{t'})^\ell f(x,y,t) g(x',y',t')\big|_{(x',y',t')=(x,y,t)}. \tag{3}$$

The bilinear equations are

$$\begin{aligned} &\left(D_x^2 + D_t + 2aD_x\right) f \cdot g = 0, \\ &\left(D_x^2 D_y + D_t D_y + 2aD_x D_y + 2\lambda D_x\right) f \cdot g = 0. \end{aligned} \tag{4}$$

Their logarithmic expansion, followed by (2), gives the continuous field equations (1). In the work of Sheng and Yu [5], we have the bilinear representation above and the continuous Gram determinant solutions given below.

For the lattice construction, we fix $\lambda = -2$. This replaces $2\lambda u$ by $-4u$ in the second field equation and $2\lambda D_x$ by $-4D_x$ in the second bilinear equation. Thus

$$\begin{aligned} &\left(D_x^2 + D_t + 2aD_x\right) f \cdot g = 0, \\ &\left(D_x^2 D_y + D_t D_y + 2aD_x D_y - 4D_x\right) f \cdot g = 0. \end{aligned} \tag{5}$$

The first relation describes the x, t dependence on each fixed-y slice, while the second connects these slices through the y derivative. We therefore consider a lattice in y, keeping x and t continuous.

The continuous Gram family is denoted by $\tau_n^{(0)}$, with the superscript (0) identifying the continuous reference family. It can be written as

$$\tau_n^{(0)} = \det_{1\le i,j\le N}\left[\delta_{ij} + \frac{\rho_i}{p_i+q_j}\left(-\frac{p_i-a}{q_j+a}\right)^n e^{\xi_i+\eta_j}\right], \qquad f = \tau_1^{(0)}, \quad g = \tau_0^{(0)}, \tag{6}$$

where

$$\xi_i = p_i x - p_i^2 t + \frac{y}{p_i - a}, \qquad \eta_j = q_j x + q_j^2 t + \frac{y}{q_j + a}. \tag{7}$$

The parameters are chosen so that the denominators are nonzero. On regions where $\tau_1^{(0)}$ and $\tau_0^{(0)}$ are positive, (2) gives smooth real physical fields. These τ functions solve the bilinear equations (5).

2.2 Bilinear semi-discrete construction and Gram determinant solutions

We seek a lattice description of the bilinear system (5) in which the transverse variable is discrete and the longitudinal variable and time remain continuous. To construct it, we start from the Hirota–Miwa equation and take a continuous limit in two lattice directions. The remaining directions will supply the pairings between neighbouring lattice functions. We then construct its Gram determinant solutions from the continuous family (6), using a standard adjacent-layer identity of the KP hierarchy.

Following the inverse-spacing convention in [15], let $T_\alpha, T_\beta, T_\gamma$ denote commuting shifts, with $\tau_\alpha = T_\alpha \tau$ and $\tau_{\alpha\beta} = T_\alpha T_\beta \tau$. The Hirota–Miwa equation [17,18] is

$$(\gamma-\beta)\tau_\alpha\tau_{\beta\gamma} + (\alpha-\gamma)\tau_\beta\tau_{\alpha\gamma} + (\beta-\alpha)\tau_\gamma\tau_{\alpha\beta} = 0. \tag{8}$$

The Hirota–Miwa equation holds for every triple of distinct directions in the multidimensional lattice. We take continuous limits in the directions labelled by α and β, keeping γ fixed. For integer coordinates ℓ_1, ℓ_2 in the first two directions, the Miwa transformation [18] is

$$\begin{aligned} &X_m = X_m^{(0)} + \frac{\ell_1}{m\alpha^m} + \frac{\ell_2}{m\beta^m}, \qquad m \geq 1, \\ &[\varepsilon] = \left(\varepsilon, \frac{\varepsilon^2}{2}, \frac{\varepsilon^3}{3}, \dots\right), \qquad T_\alpha \tau(\boldsymbol{X}) = \tau(\boldsymbol{X} + [\alpha^{-1}]). \end{aligned} \tag{9}$$

The shift T_β acts in the same way with α replaced by β. Thus a lattice step translates all hierarchy variables simultaneously. In particular,

$$\begin{aligned} &T_\alpha = \exp\left(\sum_{m\geq 1} \frac{\alpha^{-m}}{m} \partial_{X_m}\right), \\ &T_\alpha \tau = \tau + \alpha^{-1}\tau_{X_1} + \frac{\alpha^{-2}}{2}\left(\tau_{X_1X_1} + \tau_{X_2}\right) + O(\alpha^{-3}). \end{aligned} \tag{10}$$

We use the successive continuum limits described in [19, equations (1.7)–(1.9)], expressed here in the inverse-parameter convention of (8). First multiply (8) by β^{-1}, expand the β shifts using (10), and compare the coefficient of β^{-1}. The constant terms cancel, giving

$$D_{X_1}\tau_\alpha \cdot \tau_\gamma = (\alpha-\gamma)\left(\tau_\alpha\tau_\gamma - \tau\tau_{\alpha\gamma}\right). \tag{11}$$

Next let $\alpha^{-1} \to 0$. Since the shifts commute, (10) applies to both $\tau_\alpha = T_\alpha\tau$ and $\tau_{\alpha\gamma} = T_\alpha\tau_\gamma$. Substituting these expansions into (11), the terms of order α^0 cancel, and the remaining relation is

$$0 = \frac{1}{2\alpha}\left(D_{X_1}^2 - D_{X_2} + 2\gamma D_{X_1}\right)\tau\cdot\tau_\gamma + O(\alpha^{-2}). \tag{12}$$

The coefficient of α^{-1} therefore yields

$$\left(D_{X_1}^2 - D_{X_2} + 2\gamma D_{X_1}\right)\tau\cdot T_\gamma\tau = 0. \tag{13}$$

These expansions are understood as formal power series in the inverse lattice parameters. For the Gram solutions below, the shift factors are analytic near $\alpha^{-1} = \beta^{-1} = 0$. Equation (13) retains one discrete direction and two continuous variables. The higher hierarchy variables X_m, $m \geq 3$, are held fixed when identifying the DLW variables.

We now realize two retained directions using the Gram family below. On each nonconstant Gram entry, the shift T_γ has multiplier $(\gamma+q_j+a)/(\gamma-p_i+a)$. Under the spectral conditions stated below, this multiplier and its inverse have finite limits at $\gamma = 0$, giving $-(q_j+a)/(p_i-a)$. This continuation defines T_0 for this family. Choose the two retained parameters as $0, d$, with integer coordinates s, r, respectively, and set

$$n = r + s, \qquad k = -r. \tag{14}$$

Here n is the auxiliary determinant layer, and k is the transverse lattice index. Increasing s advances only n, whereas increasing r advances n and decreases k. Consequently,

$$
\begin{aligned}
&T_0\tau_n(k) = \tau_{n+1}(k), \qquad T_d\tau_n(k) = \tau_{n+1}(k-1),\\
&T_0T_d^{-1}\tau_n(k) = \tau_n(k+1).
\end{aligned}
\tag{15}
$$

The last relation identifies a forward step in the physical lattice with a composition of the retained shifts. Applying (13) in the two directions gives

$$
\begin{aligned}
&(D_{X_1}^2 - D_{X_2})\tau_n(k)\cdot\tau_{n+1}(k) = 0,\\
&(D_{X_1}^2 - D_{X_2} + 2dD_{X_1})\tau_n(k)\cdot\tau_{n+1}(k-1) = 0.
\end{aligned}
\tag{16}
$$

To identify the continuous variables with those of the DLW bilinear operator, set

$$
X_1 = x - 2bt, \qquad X_2 = -t, \qquad f_k = \tau_0(k), \quad g_k = \tau_1(k). \tag{17}
$$

The chain rule gives $D_x = D_{X_1}$ and $D_t = -2bD_{X_1} - D_{X_2}$, so that $D_{X_1}^2 - D_{X_2} = \left(D_x^2 + D_t + 2bD_x\right)$. Taking $d = -h$, with $h > 0$, converts the two relations at $n = 0$ into the following lattice.

$$
\begin{aligned}
&\left(D_x^2 + D_t + 2bD_x\right) f_k\cdot g_k = 0,\\
&\left(D_x^2 + D_t + (2b-2h)D_x\right) f_k\cdot g_{k-1} = 0.
\end{aligned}
\tag{18}
$$

The parameter b is left free at this stage. The index k labels the transverse lattice, while n labels adjacent members of the auxiliary τ sequence. We select $n = 0, 1$ without imposing a pseudo two-reduction, so the x and t flows remain independent.

We construct solutions of (18) from the continuous Gram family (6). Its transverse dependence enters through the exponential

$$
\exp\left[y\left(\frac{1}{p_i - a} + \frac{1}{q_j + a}\right)\right].
$$

On the lattice $y = kh$, we replace this exponential by the rational factor

$$
\left(\frac{1 + h/(p_i - a)}{1 - h/(q_j + a)}\right)^k. \tag{19}
$$

This factor is the multiplier of $T_0T_d^{-1}$ with $d = -h$, so it advances the physical lattice while keeping the auxiliary layer fixed. Its logarithm is $h[(p_i - a)^{-1} + (q_j + a)^{-1}] + O(h^2)$; hence (19) tends to the transverse exponential as $h \to 0$ with $kh = y$. We retain the Cauchy kernel and determinant structure of (6). The change of continuous variables (17) gives the phase $(p_i + q_j)[x - 2(b-a)t] + (q_j^2 - p_i^2)t$.

Let $N \geq 1$, $h > 0$, and let $a, b, c_{ij}, \alpha_i, \beta_j, p_i, q_j$ be real constants satisfying

$$
(p_i - a)(q_j + a)(p_i + q_j)(p_i - a + h)(q_j + a - h) \neq 0. \tag{20}
$$

For integer indices n, k, we define the common Gram family by

$$
\begin{aligned}
\tau_n^{(b)}(k) = \det_{1\leq i,j\leq N}\Bigg[&c_{ij} + \frac{\alpha_i\beta_j}{p_i + q_j}\left(-\frac{q_j + a}{p_i - a}\right)^n\\
&\times\left(\frac{1 + h/(p_i - a)}{1 - h/(q_j + a)}\right)^k \exp\big[(p_i + q_j)[x - 2(b-a)t] + (q_j^2 - p_i^2)t\big]\Bigg].
\end{aligned}
\tag{21}
$$

The auxiliary layers run in the reverse order to those in (6). With the amplitude correspondence in Appendix A, the layers $n = 0, 1$ approach the continuous pair f, g at $b = a$.

We use the Gram determinant identity of Chen et al. [20, Lemma 2.1] to verify the two pairings. In the present variables, the identity takes the following form. For constants a_i, b_j, define

$$
\begin{aligned}
\mathcal{T}_\ell(\sigma) = \det_{1\leq i,j\leq N}\Bigg[&c_{ij} + \frac{a_ib_j}{p_i + q_j}\left(-\frac{p_i + b - a - \sigma}{q_j + a - b + \sigma}\right)^\ell\\
&\times e^{(p_i+q_j)[x-2(b-a)t] + (q_j^2 - p_i^2)t}\Bigg].
\end{aligned}
\tag{22}
$$

When the displayed denominators and the numerator of the layer factor are nonzero, the adjacent determinants satisfy

$$\left(D_x^2 + D_t + 2\sigma D_x\right) \mathcal{T}_{\ell+1}(\sigma) \cdot \mathcal{T}_\ell(\sigma) = 0. \tag{23}$$

The correspondence with the cited identity, including its auxiliary variable, is given in Appendix A. The amplitudes a_i, b_j may depend on a fixed lattice site.

Theorem 2.1. Under (20), let $\tau_n^{(b)}(k)$ be defined by (21). Then $f_k = \tau_0^{(b)}(k)$ and $g_k = \tau_1^{(b)}(k)$ satisfy the two bilinear equations (18).

Proof. We apply (23) at a fixed lattice site k, with

$$a_i(k) = \alpha_i \left(1 + \frac{h}{p_i - a}\right)^k, \qquad b_j(k) = \beta_j \left(1 - \frac{h}{q_j + a}\right)^{-k}. \tag{24}$$

With (24), the zero-layer phase and amplitude agree with (21). Write the corresponding determinant as $\mathcal{T}_\ell(k;\sigma)$. At $\sigma = b$, its inverse layer multiplier is $-(q_j + a)/(p_i - a)$, and hence

$$\mathcal{T}_0(k;b) = \tau_0^{(b)}(k) = f_k, \qquad \mathcal{T}_{-1}(k;b) = \tau_1^{(b)}(k) = g_k. \tag{25}$$

Taking $\ell = -1$ and $\sigma = b$ in (23) gives $\left(D_x^2 + D_t + 2bD_x\right) f_k \cdot g_k = 0$.

For the second pairing, the zero layer is independent of σ, so $\mathcal{T}_0(k; b-h) = f_k$. The inverse layer factor at $\sigma = b - h$ obeys

$$-\frac{q_j + a - h}{p_i - a + h} \left(\frac{1 + h/(p_i - a)}{1 - h/(q_j + a)}\right)^k = -\frac{q_j + a}{p_i - a} \left(\frac{1 + h/(p_i - a)}{1 - h/(q_j + a)}\right)^{k-1}. \tag{26}$$

Thus the change from $\sigma = b$ to $\sigma = b - h$ in the inverse layer absorbs one backward lattice step. All other entry factors agree, giving $\mathcal{T}_{-1}(k; b-h) = \tau_1^{(b)}(k-1) = g_{k-1}$. Applying (23) with $\ell = -1$ and $\sigma = b - h$ therefore gives $\left(D_x^2 + D_t + (2b - 2h)D_x\right) f_k \cdot g_{k-1} = 0$, proving the theorem. □

2.3 Endpoint and centered semi-discretizations

The lattice (18) and its Gram family (21) depend on the free parameter b. We now compare them with the continuous DLW system at fixed a. Choosing b and assigning the two τ functions to transverse positions gives the endpoint and centered formulations.

Definition 2.1 (Endpoint-SD). Set $b = a$ and place both f_k^{E} and g_k^{E} at $y_k = kh$. This gives the endpoint-based semi-discretization, abbreviated Endpoint-SD,

$$\begin{aligned} &\left(D_x^2 + D_t + 2aD_x\right) f_k^{\mathrm{E}} \cdot g_k^{\mathrm{E}} = 0, \\ &\left(D_x^2 + D_t + (2a - 2h)D_x\right) f_k^{\mathrm{E}} \cdot g_{k-1}^{\mathrm{E}} = 0. \end{aligned} \tag{27}$$

The endpoint tau functions are obtained from the determinant family through

$$f_k^{\mathrm{E}} = \tau_0^{\mathrm{E}}(k), \qquad g_k^{\mathrm{E}} = \tau_1^{\mathrm{E}}(k). \tag{28}$$

Substitution of $b = a$ in (21) gives

$$\begin{aligned} \tau_n^{\mathrm{E}}(k) = \det_{1 \le i,j \le N} \Bigg[& c_{ij} + \frac{\alpha_i \beta_j}{p_i + q_j} \left(-\frac{q_j + a}{p_i - a}\right)^n \\ & \times \left(\frac{1 + h/(p_i - a)}{1 - h/(q_j + a)}\right)^k \exp\left[(p_i + q_j)x + (q_j^2 - p_i^2)t\right] \Bigg]. \end{aligned} \tag{29}$$

By Theorem 2.1, the functions in (28) satisfy (27).

To examine consistency, define

$$\Phi(s) = \left(D_x^2 + D_t + 2(a + s)D_x\right) f(x,y,t) \cdot g(x, y+s, t). \tag{30}$$

The endpoint pairings correspond to $\Phi(0)$ and $\Phi(-h)$.

Proposition 2.1. *For sufficiently smooth f, g,*

$$\begin{aligned}\frac{\Phi(0)-\Phi(-h)}{h} &= \left(D_x^2+D_t+2aD_x\right) f\cdot g_y + 2D_x f\cdot g \\ &\quad -\frac{h}{2}\left[\left(D_x^2+D_t+2aD_x\right) f\cdot g_{yy} + 4D_x f\cdot g_y\right] + O(h^2).\end{aligned} \tag{31}$$

Proof. Differentiation gives $\Phi'(0) = \left(D_x^2+D_t+2aD_x\right) f\cdot g_y + 2D_x f\cdot g$ and $\Phi''(0) = \left(D_x^2+D_t+2aD_x\right) f\cdot g_{yy} + 4D_x f\cdot g_y$. Taylor expansion proves (31). For solutions of (5), ordinary differentiation of the first relation gives $\left(D_x^2+D_t+2aD_x\right) f_y\cdot g + \left(D_x^2+D_t+2aD_x\right) f\cdot g_y = 0$, while the second gives $\left(D_x^2+D_t+2aD_x\right) f_y\cdot g - \left(D_x^2+D_t+2aD_x\right) f\cdot g_y - 4D_x f\cdot g = 0$. Combining these identities yields $\Phi'(0)=0$. □

The first-order term in (31) arises from the one-sided placement of the endpoint pairings. A symmetric arrangement is obtained by choosing

$$b = a + \frac{h}{2}. \tag{32}$$

Definition 2.2 (Centered-SD). Set $b = a + h/2$. Place g_k^{C} at $y_k = kh$ and f_k^{C} at $y_{k-\frac{1}{2}} = (k-\frac{1}{2})h$. The centered semi-discretization, abbreviated Centered-SD, is the bilinear system

$$\begin{aligned}\left(D_x^2+D_t+(2a-h)D_x\right) f_k^{\mathrm{C}}\cdot g_{k-1}^{\mathrm{C}} &= 0,\\ \left(D_x^2+D_t+(2a+h)D_x\right) f_k^{\mathrm{C}}\cdot g_k^{\mathrm{C}} &= 0.\end{aligned} \tag{33}$$

The two neighbouring functions $g_{k-1}^{\mathrm{C}}, g_k^{\mathrm{C}}$ are symmetric about f_k^{C}, and the two operator parameters are symmetric about a.

Substituting $b = a + h/2$ into (21) gives

$$\begin{aligned}\tau_n^{\mathrm{C}}(k) = \det_{1\le i,j\le N}\Bigg[c_{ij} + \frac{\alpha_i\beta_j}{p_i+q_j}\left(-\frac{q_j+a}{p_i-a}\right)^n \\ \times\left(\frac{1+h/(p_i-a)}{1-h/(q_j+a)}\right)^k \exp\big[(p_i+q_j)(x-ht)+(q_j^2-p_i^2)t\big]\Bigg].\end{aligned} \tag{34}$$

The functions $f_k^{\mathrm{C}} = \tau_0^{\mathrm{C}}(k)$, $g_k^{\mathrm{C}} = \tau_1^{\mathrm{C}}(k)$ satisfy (33) by Theorem 2.1. Both formulations therefore use the same Gram kernel and lattice factor; the change of b is recorded in their continuous phases.

At fixed h, (33) follows exactly from (18) through (32). For the continuous comparison, a remains fixed. Using Φ from (30) at the location of f_k^{C}, the two pairings correspond to $\Phi(-h/2)$ and $\Phi(h/2)$.

Proposition 2.2. *For sufficiently smooth f, g,*

$$\begin{aligned}\frac{\Phi(h/2)+\Phi(-h/2)}{2} &= \left(D_x^2+D_t+2aD_x\right) f\cdot g + O(h^2),\\ \frac{\Phi(h/2)-\Phi(-h/2)}{h} &= \left(D_x^2+D_t+2aD_x\right) f\cdot g_y + 2D_x f\cdot g + O(h^2).\end{aligned} \tag{35}$$

Proof. Taylor expansion gives $\Phi(\pm h/2) = \Phi(0) \pm (h/2)\Phi'(0) + (h^2/8)\Phi''(0) + O(h^3)$. Taking the sum and difference proves (35). As in the proof of Proposition 2.1, the limiting relations are equivalent to the continuous bilinear pair (5). □

The endpoint and symmetric pairings give first- and second-order bilinear consistency, respectively. To reconstruct smooth real physical fields through logarithmic derivatives, we use a positive parameter class for these determinant solutions.

Proposition 2.3. *Choose real parameters satisfying $c_{ij} = \delta_{ij}$, $\alpha_i < 0$, and $\beta_i > 0$. If*

$$-a < q_N < \cdots < q_1 < -p_1 < \cdots < -p_N, \qquad 0 < h < q_N + a, \tag{36}$$

then both determinant families are positive for every integer n, k and finite real x, t.

Proof. The nonconstant part of each determinant is a product of positive diagonal matrices and the Cauchy matrix with entries $-1/(p_i+q_j)$. Indeed, $-(q_j+a)/(p_i-a)>0$ and $(1+h/(p_i-a))/(1-h/(q_j+a))>0$, and these factors separate into row and column factors. Every nonempty principal minor of this Cauchy matrix satisfies

$$\det_{i,j\in I}\left[-\frac{1}{p_i+q_j}\right]=\frac{\prod_{i<j,\ i,j\in I}(p_i-p_j)(q_i-q_j)}{\prod_{i,j\in I}(-p_i-q_j)}>0. \tag{37}$$

Positive diagonal multiplication preserves these signs. Expanding each determinant as the sum of the principal minors of its nonconstant part, including the empty minor 1, proves positivity. The additional Centered-SD phase also consists of positive row and column factors. □

The common bilinear lattice and its determinant identity thus support both parameter choices. On the positive class, the resulting determinants yield smooth real physical fields. Their derivative identities are collected in Appendix A.

3 Integrable nonlinear semi-discrete formulations

The nonlinear Endpoint-SD system is given by (41)–(42), and the Centered-SD system by (49), closed by the reconstruction (50). Their Darboux–Lax pairs are (56) and (57). We first state these systems in the physical fields and then derive them from the bilinear equations. Section 3.4 examines their continuous limits and the approximation of the exact wave fields.

3.1 The endpoint-based semi-discretization (Endpoint-SD)

We reconstruct the endpoint fields from the logarithmic ratio and product of $f_k^{\mathrm{E}}, g_k^{\mathrm{E}}$. For a lattice function $w_k(x,t)$, define

$$\Delta_h w_k=\frac{w_k-w_{k-1}}{h}. \tag{38}$$

The Endpoint-SD bilinear equations are

$$\begin{aligned}&\left(D_x^2+D_t+2aD_x\right)f_k^{\mathrm{E}}\cdot g_k^{\mathrm{E}}=0,\\&\left(D_x^2+D_t+(2a-2h)D_x\right)f_k^{\mathrm{E}}\cdot g_{k-1}^{\mathrm{E}}=0.\end{aligned} \tag{39}$$

We define the physical fields by

$$u_k=2\partial_x\log\frac{f_k^{\mathrm{E}}}{g_k^{\mathrm{E}}},\qquad v_k=\frac{2}{h}\partial_x\log\frac{f_k^{\mathrm{E}}g_k^{\mathrm{E}}}{f_{k-1}^{\mathrm{E}}g_{k-1}^{\mathrm{E}}}. \tag{40}$$

The field u_k is associated with the node $y_k=kh$, while v_k lies at the edge midpoint $y_{k-\frac{1}{2}}$. These locations are used when comparing the lattice fields with their continuous counterparts.

Proposition 3.1. *Let $f_k^{\mathrm{E}}, g_k^{\mathrm{E}}$ be positive, sufficiently smooth solutions of (39). The physical fields (40) satisfy the Endpoint-SD system*

$$\Delta_h u_{k,t}=-\partial_x\Delta_h\left(\frac{1}{2}u_k^2+2au_k\right)-v_{k,xx}, \tag{41}$$

$$\begin{aligned}v_{k,t}=&-\partial_x^2\Delta_h u_k\\&-\partial_x\left[\left(\frac{u_k+u_{k-1}}{2}+2a-2h\right)v_k+\frac{h}{4}v_k^2\right.\\&\qquad\left.-2(u_k+u_{k-1})-\frac{h}{4}(\Delta_h u_k)^2\right].\end{aligned} \tag{42}$$

Proof. For either tau family, use the two logarithmic ratios

$$r_k = \log \frac{f_k}{g_k}, \qquad q_k = \log \frac{f_k}{g_{k-1}}. \tag{43}$$

For the common bilinear lattice (18), division by the corresponding tau products gives

$$\begin{aligned} r_{k,t} &= r_{k,xx} - r_{k,x}^2 - 2br_{k,x} - 2\partial_x^2 \log f_k, \\ q_{k,t} &= q_{k,xx} - q_{k,x}^2 - 2(b-h)q_{k,x} - 2\partial_x^2 \log f_k. \end{aligned} \tag{44}$$

Here we used $\log(f_k g_k) = 2\log f_k - r_k$ and $\log(f_k g_{k-1}) = 2\log f_k - q_k$. The identities (44) will also be used for the centered formulation and the Lax representation.

For Endpoint-SD, set $b = a$ and take $(f_k, g_k) = (f_k^{\mathrm{E}}, g_k^{\mathrm{E}})$. The field definitions imply

$$u_k = 2r_{k,x}, \qquad q_{k,x} = \frac{u_k + u_{k-1}}{4} + \frac{h}{4} v_k, \qquad 4\Delta_h \partial_x \log f_k = v_k + \Delta_h u_k. \tag{45}$$

Applying $2\partial_x$ to the first equation of (44) gives

$$u_{k,t} = u_{k,xx} - \partial_x \left(\frac{1}{2} u_k^2 + 2a u_k \right) - 4\partial_x^3 \log f_k. \tag{46}$$

Taking its lattice difference and using the last identity in (45) yields (41).

For the second field equation, subtract the average of the first relation in (44) and its backward shift from the second relation. Use (45), multiply by $4/h$, and differentiate in x to obtain (42). The algebraic identities are given in Appendix B. □

Equations (41)–(42) describe the endpoint fields reconstructed from the bilinear lattice. Applying (40) to the positive Gram determinants of Section 2.3 gives smooth exact fields, which provide reference solutions for the analysis and numerical experiments below.

3.2 The centered semi-discretization (Centered-SD)

The centered construction uses the symmetric placement of f_k^{C} between g_{k-1}^{C} and g_k^{C}. Its bilinear pairings are

$$\begin{aligned} \left(D_x^2 + D_t + (2a+h)D_x\right) f_k^{\mathrm{C}} \cdot g_k^{\mathrm{C}} = 0, \\ \left(D_x^2 + D_t + (2a-h)D_x\right) f_k^{\mathrm{C}} \cdot g_{k-1}^{\mathrm{C}} = 0. \end{aligned} \tag{47}$$

With f_k^{C} located at $y_k = (k-1/2)h$, define

$$u_k = \partial_x \log \frac{\left(f_k^{\mathrm{C}}\right)^2}{g_{k-1}^{\mathrm{C}} g_k^{\mathrm{C}}}, \qquad v_k = \partial_x \Delta_h \log\left(\left(f_k^{\mathrm{C}}\right)^2 g_{k-1}^{\mathrm{C}} g_k^{\mathrm{C}}\right). \tag{48}$$

The field u_k is nodal, and v_k lies at the edge midpoint $y_k - h/2$. For these exact tau fields, the logarithmic difference is

$$\omega_k = \partial_x \log \frac{g_k^{\mathrm{C}}}{g_{k-1}^{\mathrm{C}}}.$$

Proposition 3.2. *Let $f_k^{\mathrm{C}}, g_k^{\mathrm{C}}$ be positive, sufficiently smooth solutions of (47). The physical fields (48) satisfy the Centered-SD system*

$$
\begin{aligned}
\Delta_h u_{k,t} &= -\partial_x \Delta_h \left[\frac{1}{2} u_k^2 + 2a u_k + \frac{\omega_k^2}{2} - h\omega_k \right] - v_{k,xx}, \\
v_{k,t} &= -\partial_x^2 \Delta_h u_k \\
&\quad - \partial_x \Bigg\{ \Delta_h \left[\frac{1}{2} u_k^2 + 2a u_k + \frac{\omega_k^2}{2} - h\omega_k \right] \\
&\qquad + \frac{2}{h} \left[(u_k + 2a)\omega_k + (u_{k-1} + 2a)\omega_{k-1} - h(u_k + u_{k-1}) \right] \Bigg\}.
\end{aligned}
\tag{49}
$$

Here ω_k is reconstructed from the physical fields by

$$
\omega_k = \frac{h}{2}(v_k - \Delta_h u_k) - \omega_{k-1}, \qquad k = 1, \dots, N. \tag{50}
$$

with the compatible lower trace $\omega_0 = \partial_x \log(g_0^{\mathrm{C}} / g_{-1}^{\mathrm{C}})$. On a finite lattice, the recurrence uniquely determines ω from u, v and this trace.

Appendix B gives the closed form of the reconstruction and the conditions on the lower trace.

Proof. Use the ratios (43) with $(f_k, g_k) = (f_k^{\mathrm{C}}, g_k^{\mathrm{C}})$. Their derivatives satisfy

$$
u_k = r_{k,x} + q_{k,x}, \qquad (q_k - r_k)_x = \partial_x \log \frac{g_k}{g_{k-1}}. \tag{51}
$$

The field definitions also give

$$
v_k - \Delta_h u_k = \frac{2}{h} \partial_x \log \frac{g_k}{g_{k-2}} = \frac{2}{h} \left[\partial_x \log \frac{g_k}{g_{k-1}} + \partial_x \log \frac{g_{k-1}}{g_{k-2}} \right]. \tag{52}
$$

By the recurrence (50) and its specified initial value, $\omega_k = (q_k - r_k)_x = \partial_x \log(g_k / g_{k-1})$. Hence $r_{k,x} = (u_k - \omega_k)/2$ and $q_{k,x} = (u_k + \omega_k)/2$. Set $b = a + h/2$ in (44). Subtracting its first relation from the second and differentiating in x gives

$$
\omega_{k,t} = \omega_{k,xx} - \partial_x [(u_k + 2a)\omega_k - h u_k]. \tag{53}
$$

Adding the two relations in (44) uses $r_{k,x}^2 + q_{k,x}^2 = (u_k^2 + \omega_k^2)/2$. Apply $\partial_x \Delta_h$ to the sum. Since (48) implies $4\Delta_h \partial_x \log f_k = v_k + \Delta_h u_k$, the second derivatives combine to give the first equation in (49).

Differentiating $v_k = \Delta_h u_k + 2(\omega_k + \omega_{k-1})/h$ in t and twice in x gives

$$
\begin{aligned}
v_{k,t} &= \Delta_h u_{k,t} + \frac{2}{h}(\omega_{k,t} + \omega_{k-1,t}), \\
v_{k,xx} &= \partial_x^2 \Delta_h u_k + \frac{2}{h}(\omega_{k,xx} + \omega_{k-1,xx}).
\end{aligned}
\tag{54}
$$

Substitution of (53) and the first equation of (49), followed by the second identity in (54), gives the second equation of (49). □

3.3 Darboux–Lax representations

The two tau pairings in (18) also give compatible linear equations. Assume that the tau functions are positive, sufficiently smooth solutions of (18), and use the logarithmic ratios r_k, q_k in (43). For Endpoint-SD take $b = a$; for Centered-SD take $b = a + h/2$, with the compatible lower trace in (50). Define the operators on a sequence $\psi = (\psi_k)$ by

$$
\begin{aligned}
(L\psi)_k &= (\partial_x + r_{k-1,x})\psi_{k-1} - (\partial_x + q_{k,x} - h)\psi_k, \\
(M\psi)_k &= (\partial_x^2 - 2b\partial_x + 2\partial_x^2 \log f_k)\psi_k.
\end{aligned}
\tag{55}
$$

Thus L relates neighbouring lattice sites, while M governs the time evolution at each site. In the physical variables these operators are as follows.

Endpoint-SD. The relations (45) give

$$\begin{aligned}(L^{\mathrm{E}}\psi)_k &= \left(\partial_x + \frac{u_{k-1}}{2}\right)\psi_{k-1} - \left(\partial_x + \frac{u_k + u_{k-1} + hv_k}{4} - h\right)\psi_k,\\ (M^{\mathrm{E}}\psi)_k &= \left(\partial_x^2 - 2a\partial_x + 2\partial_x^2 \log f_k\right)\psi_k.\end{aligned} \tag{56}$$

Centered-SD. The centered field definitions give

$$\begin{aligned}(L^{\mathrm{C}}\psi)_k &= \left(\partial_x + \frac{u_{k-1} - \omega_{k-1}}{2}\right)\psi_{k-1} - \left(\partial_x + \frac{u_k + \omega_k}{2} - h\right)\psi_k,\\ (M^{\mathrm{C}}\psi)_k &= \left(\partial_x^2 - (2a+h)\partial_x + 2\partial_x^2 \log f_k\right)\psi_k.\end{aligned} \tag{57}$$

Write $[L, M] = LM - ML$. The following proposition gives the compatibility relation for the auxiliary equations $L\psi = 0$, $\psi_t = M\psi$.

Proposition 3.3. *The operators (56)–(57) satisfy, componentwise,*

$$(L^{\mathrm{E}})_t + [L^{\mathrm{E}}, M^{\mathrm{E}}] = -\frac{1}{2}\partial_x(u_k + u_{k-1} + hv_k)L^{\mathrm{E}}, \tag{58}$$

$$(L^{\mathrm{C}})_t + [L^{\mathrm{C}}, M^{\mathrm{C}}] = -\partial_x(u_k + \omega_k)L^{\mathrm{C}}. \tag{59}$$

Proof. The first bilinear equation at site $k-1$ and the second at site k give two solutions of the time equation at the respective sites,

$$\begin{aligned}\partial_t\left(\frac{g_{k-1}}{f_{k-1}}\right) &= \left(\partial_x^2 - 2b\partial_x + 2\partial_x^2 \log f_{k-1}\right)\frac{g_{k-1}}{f_{k-1}},\\ \partial_t\left(e^{hx+(h^2-2bh)t}\frac{g_{k-1}}{f_k}\right) &= \left(\partial_x^2 - 2b\partial_x + 2\partial_x^2 \log f_k\right)\left(e^{hx+(h^2-2bh)t}\frac{g_{k-1}}{f_k}\right).\end{aligned} \tag{60}$$

Their logarithmic derivatives give the Darboux factors $\partial_x + r_{k-1,x}$ and $\partial_x + q_{k,x} - h$ appearing in L. Both factors transform their time operators to the same operator, because

$$\begin{aligned}2\partial_x^2 \log f_{k-1} - 2r_{k-1,xx} &= 2\partial_x^2 \log g_{k-1},\\ 2\partial_x^2 \log f_k - 2q_{k,xx} &= 2\partial_x^2 \log g_{k-1}.\end{aligned} \tag{61}$$

The Darboux intertwining relations therefore imply

$$\big((L_t + [L, M])\psi\big)_k = -2q_{k,xx}(L\psi)_k. \tag{62}$$

Appendix C gives the component calculation directly from (44). Substituting the physical field definitions into $q_{k,x}$ proves (58)–(59). Along $\psi_t = M\psi$, (62) becomes

$$\partial_t(L\psi)_k = \left(\partial_x^2 - 2b\partial_x + 2\partial_x^2 \log g_{k-1}\right)(L\psi)_k. \tag{63}$$

Hence the time evolution preserves the lattice equation $L\psi = 0$.

The coefficient relations also recover the nonlinear field equations. For Endpoint-SD they give

$$\begin{aligned}u_{k,t} &= u_{k,xx} - (u_k + 2a)u_{k,x} - 4\partial_x^3 \log f_k,\\ \Delta_h(2\partial_x^2 \log f_k) &= \frac{1}{2}\partial_x(v_k + \Delta_h u_k).\end{aligned} \tag{64}$$

For Centered-SD the corresponding evolution is

$$u_{k,t} = u_{k,xx} - \partial_x\left[\frac{1}{2}u_k^2 + 2au_k + \frac{1}{2}\omega_k^2 - h\omega_k\right] - 4\partial_x^3 \log f_k. \tag{65}$$

Taking lattice differences and using the field reconstructions yields (41)–(42) and (49). The substitutions for the second fields are recorded in Appendix C. □

3.4 Continuous limit and approximation of the physical fields

We compare the equations at the edge midpoint of each formulation. The physical fields u_k, v_k occupy their respective node and edge locations, with the centered staggering specified in Section 3.2. For Centered-SD, write

$$W_k = \frac{4\omega_k}{h}, \qquad v_k = \Delta_h u_k + \frac{W_k + W_{k-1}}{2}.$$

The smooth reconstruction assumption below is stated in terms of W.

Proposition 3.4. *On a fixed compact region, suppose the lattice fields admit smooth reconstructions $u(x,y,t), v(x,y,t)$, with the derivatives required below bounded uniformly as $h \to 0$.*
Endpoint-SD. *For solutions of (41)–(42), the reconstructed fields satisfy*

$$\begin{aligned} u_{yt} &= -v_{xx} - \partial_x[(u+2a)u_y] + O(h^2), \\ v_t &= -u_{xxy} - \partial_x[(u+2a)v - 4u] \\ &\quad - \frac{h}{4}\partial_x(v^2 - u_y^2 - 8v) + O(h^2). \end{aligned} \tag{66}$$

Centered-SD. *For solutions of (49), assume the recurrence (50) selects a smooth reconstructed family with $W = v - u_y + O(h^2)$, including the derivative bounds required in the equations. Then*

$$\begin{aligned} u_{yt} &= -v_{xx} - \partial_x[(u+2a)u_y] + O(h^2), \\ v_t &= -u_{xxy} - \partial_x[(u+2a)v - 4u] + O(h^2). \end{aligned} \tag{67}$$

All continuous quantities are evaluated at the corresponding edge midpoint. Under these smoothness and reconstruction assumptions, sampling a solution of the continuous DLW system gives defects of orders $O(h^2)$ and $O(h)$ in the two Endpoint-SD equations, and $O(h^2)$ in both Centered-SD equations.

Proof. Midpoint expansion gives $\Delta_h u = u_y + h^2 u_{yyy}/24 + O(h^4)$ and $\frac{u(y+h/2)+u(y-h/2)}{2} = u + h^2 u_{yy}/8 + O(h^4)$. The first Endpoint-SD equation therefore gives the first relation in (66). In its second equation, the flux becomes

$$(u+2a)v - 4u + u_{xy} + \frac{h}{4}(v^2 - u_y^2 - 8v) + O(h^2), \tag{68}$$

which gives the second relation in (66) after differentiation in x.

For Centered-SD, the reconstruction has the expansion

$$v = u_y + W + h^2\left(\frac{u_{yyy}}{24} + \frac{W_{yy}}{8}\right) + O(h^4). \tag{69}$$

The extra flux in its first equation is $\omega^2/2 - h\omega = h^2(W^2/32 - W/4)$. On the stated smooth family, its lattice difference and subsequent x derivative are $O(h^2)$, which gives the first relation in (67).

Scaling (53) gives $W_t = W_{xx} - \partial_x[(u+2a)W - 4u]$. Differentiating the reconstruction in time and using this equation, the right-hand side for v_t is

$$\begin{aligned} &- \partial_y\partial_x\left(\frac{1}{2}u^2 + 2au\right) - v_{xx} + (v - u_y)_{xx} - \partial_x[(u+2a)(v-u_y) - 4u] + O(h^2) \\ &= -u_{xxy} - \partial_x[(u+2a)v - 4u] + O(h^2). \end{aligned} \tag{70}$$

This proves (67). □

The smooth-family assumption includes the lower trace in the reconstruction (50), together with the resulting spatial derivatives. For matched exact waves, this expansion follows from the Gram formulas.

Fix the spectral parameters p_i, q_j of the continuous family (6)–(7). Their combined phase is $(p_i + q_j)x + (q_j^2 - p_i^2)t + [(p_i - a)^{-1} + (q_j + a)^{-1}]y$. To match this family, evaluate the Centered-SD Gram formula (34) at

the shifted parameters $p_i - h/2$ and $q_j + h/2$, with $c_{ij} = \delta_{ij}$ and $\alpha_i\beta_j = -\rho_i((p_i - a) - h/2)/((q_j + a) + h/2)$. The x, t phase then agrees exactly with the continuous one, and the y phase rate is

$$\frac{1}{h}\log\frac{(p_i - a + h/2)(q_j + a + h/2)}{(p_i - a - h/2)(q_j + a - h/2)} = \frac{1}{p_i - a} + \frac{1}{q_j + a} + O(h^2). \tag{71}$$

At their respective staggered positions, the g amplitude agrees with the continuous amplitude, while the f amplitude differs by the factor $\sqrt{(1 - h^2/(4(p_i - a)^2))/(1 - h^2/(4(q_j + a)^2))} = 1 + O(h^2)$. For nonzero spectral denominators, sufficiently small h, and positive tau functions bounded away from zero on the fixed region, the determinants and their required derivatives therefore differ from the continuous counterparts by $O(h^2)$. Symmetric averaging in (48) gives $u_h = u + O(h^2)$; applying the midpoint difference to the logarithmic potential gives $v_h = v + O(h^2)$. Thus both exact physical fields approximate their continuous counterparts at second order.

The extra flux also distinguishes the initial derivative errors of the two semi-discrete formulations. With common continuous physical samples and the same lower value of u_t, subtracting their first equations and summing along the lattice gives

$$(u_{k,t})_{\mathrm{C}} - (u_{k,t})_{\mathrm{E}} = -\partial_x\left[\left(\frac{\omega_k^2}{2} - h\omega_k\right) - \left(\frac{\omega_0^2}{2} - h\omega_0\right)\right]. \tag{72}$$

On the smooth reconstructed family, this difference is $O(h^2)$. The two formulations therefore share the second-order approximation of the first equation while retaining different leading derivative errors. This identity connects the nonlinear flux to the initial spatial comparisons in Section 5.2. The midpoint expansions behind (66) and (67) are given in Appendix C.

4 Numerical formulation

In this section, we develop numerical formulations for computing the physical fields introduced above. The semi-discrete equations retain continuous dependence on x and t, so their numerical evolution requires a further spatial approximation, initial and boundary data, and a time-integration scheme. We specify these ingredients for Endpoint-SD and Centered-SD, together with a direct finite-difference discretization of the continuous DLW system as a baseline. Section 4.1 introduces the common differentiation operators in x and the resulting spatial formulations. Section 4.2 specifies the initial and boundary data and describes how the fields are advanced in time. In Section 4.3, we use the Centered-SD conservation law to construct a shared moving mesh, allowing the longitudinal resolution to adapt to the evolving fields.

4.1 Fourier differentiation and spatial formulations

We first discretize the remaining x derivatives to obtain systems that can be advanced in time. Endpoint-SD, Centered-SD, and the direct finite-difference baseline use the same Fourier differentiation in x, providing a common longitudinal approximation for examining their different transverse formulations. Write the periodic interval as $[-L, L)$, where L is its half-width and the period is $2L$. On Fourier mode m, the numerical operators $\widetilde{\partial}_x$ and $\widetilde{\partial}_{xx}$ have multipliers $\mathrm{i}\pi m/L$ and $-(\pi m/L)^2$, respectively. The first-derivative multiplier is set to zero at the Nyquist frequency, and nonlinear products are evaluated at the grid points. Throughout the numerical formulation, the tilde denotes the numerical derivative in the physical coordinate x. This subsection uses Fourier differentiation. Section 4.3 specifies the mapped three-point realization used for both meshes in the fixed-versus-adaptive comparison.

Both semi-discrete formulations place u at nodes and v at edge midpoints; FD places both fields at nodes. We use the backward transverse difference (38) and write adjacent averages and centered transverse differences explicitly. Recalling the physical-field equations (41)–(42), Endpoint-SD takes the form

$$
\begin{aligned}
\Delta_h \dot{u}_k &= -\widetilde{\partial}_x \Delta_h \left(\frac{1}{2} u_k^2 + 2a u_k \right) - \widetilde{\partial}_{xx} v_k, \\
\dot{v}_k &= -\widetilde{\partial}_x \Bigg[\left(\frac{u_k + u_{k-1}}{2} + 2a - 2h \right) v_k + \frac{h}{4} v_k^2 \\
&\qquad - 2(u_k + u_{k-1}) - \frac{h}{4} (\Delta_h u_k)^2 \Bigg] - \widetilde{\partial}_{xx} \Delta_h u_k.
\end{aligned}
\tag{73}
$$

For Centered-SD, the field definition in Section 3.2 gives $v_k = \Delta_h u_k + 2(\omega_k + \omega_{k-1})/h$. We retain this relation in the numerical formulation. The equations for the time derivatives are

$$
\begin{aligned}
\Delta_h \dot{u}_k &= -\widetilde{\partial}_x \Delta_h [\frac{1}{2} u_k^2 + 2a u_k + \omega_k^2/2 - h\omega_k] - \widetilde{\partial}_{xx} v_k, \\
\dot{\omega}_k &= \widetilde{\partial}_{xx} \omega_k - \widetilde{\partial}_x [(u_k + 2a)\omega_k - h u_k], \\
\dot{v}_k &= \Delta_h \dot{u}_k + \frac{2}{h} (\dot{\omega}_k + \dot{\omega}_{k-1}).
\end{aligned}
\tag{74}
$$

The first equation is summed from the prescribed lower value of $\dot{u}$. Advancing (u, ω) and recovering v through the field relation implements the physical-field equations of Proposition 3.2. The same differentiated relation supplies $\dot{v}$ when measuring the initial spatial error.

For comparison, direct discretization of the continuous DLW system gives the FD baseline,

$$
\begin{aligned}
\Delta_h \dot{u}_k &= -\widetilde{\partial}_x \Delta_h \left(\frac{1}{2} u_k^2 + 2a u_k \right) - \widetilde{\partial}_{xx} \frac{v_k + v_{k-1}}{2}, \\
\dot{v}_k &= -\widetilde{\partial}_x [(u_k + 2a) v_k - 4u_k] - \widetilde{\partial}_{xx} \frac{u_{k+1} - u_{k-1}}{2h}.
\end{aligned}
\tag{75}
$$

The three spatial formulations use the same Fourier differentiation in x and the prescribed lower-boundary data for recovering the u time derivative. Their different approximations enter through the transverse differences, nonlinear fluxes, and placement of the physical fields. This common numerical setting allows us to examine how the endpoint and centered lattice constructions affect the computed waves, with direct finite differences providing a reference discretization of the continuous equations. It also connects the consistency analysis of Section 3 with the spatial errors measured in the numerical experiments.

4.2 Initial and boundary data and time integration

In this subsection, we specify the initial and boundary data and describe how the resulting spatial systems are advanced in time. The initial states are obtained by sampling the same continuous reference fields at the native locations of each formulation. The state advanced in time is

$$
\boldsymbol{z} = \begin{cases} (u_k, v_k)_k, & \text{Endpoint-SD and FD,} \\ (u_k, \omega_k)_k, & \text{Centered-SD.} \end{cases}
$$

Each component includes its values at all longitudinal nodes. Centered-SD reconstructs v through the relation in Section 4.1. At each stage, the current state and the prescribed boundary data supply the time derivatives required by the integrator. We use forward Euler, classical RK4, and Crank–Nicolson to examine how different temporal approximations affect the computed physical fields.

For Centered-SD, the initial lower auxiliary value is $\omega_0 = h(v^* - u_y^*)/4$, evaluated at the lower node. The remaining values follow from the linear reconstruction (50),

$$
\omega_k = \frac{h}{2} (v_k - \Delta_h u_k) - \omega_{k-1}. \tag{76}
$$

This recurrence reproduces the prescribed physical field v on every internal edge. During evolution, the lower auxiliary value follows the same ω equation as the interior values. The lower value of u_t is supplied by the continuous reference solution at every stage.

Endpoint-SD uses the prescribed lower derivative and the continuous lower ghost value of u in its backward lattice relations. For FD, the lower data are supplied in the same way, while the upper ghost value extrapolates the departure from the continuous reference. Writing $e_k = u_k - u_k^*$, this closure is $u_{N+1} = u_{N+1}^* + 3e_N - 3e_{N-1} + e_{N-2}$. The reference values are evaluated at the current stage time. Fourier differentiation acts on a periodic x interval extending beyond the principal wave profiles; the products are evaluated at the nodes without filtering, dealiasing or artificial viscosity.

Let $t_n = n\Delta t$, and let $\boldsymbol{z}^n$ denote the state at t_n. Writing the evolution equations as $\dot{\boldsymbol{z}} = F(t, \boldsymbol{z})$, forward Euler advances the state by $\boldsymbol{z}^{n+1} = \boldsymbol{z}^n + \Delta t F(t_n, \boldsymbol{z}^n)$. Classical RK4 uses four stage derivatives,

$$\begin{aligned} K_1 &= F(t_n, \boldsymbol{z}^n), & K_2 &= F(t_n + \Delta t/2, \boldsymbol{z}^n + \Delta t K_1/2), \\ K_3 &= F(t_n + \Delta t/2, \boldsymbol{z}^n + \Delta t K_2/2), & K_4 &= F(t_n + \Delta t, \boldsymbol{z}^n + \Delta t K_3), \\ \boldsymbol{z}^{n+1} &= \boldsymbol{z}^n + \frac{\Delta t}{6}(K_1 + 2K_2 + 2K_3 + K_4). \end{aligned} \tag{77}$$

Crank–Nicolson determines the next state from

$$\boldsymbol{z}^{n+1} = \boldsymbol{z}^n + \frac{\Delta t}{2}\big[F(t_n, \boldsymbol{z}^n) + F(t_{n+1}, \boldsymbol{z}^{n+1})\big]. \tag{78}$$

We solve the implicit step by fixed-point iteration from an Euler prediction, using the scaled state (u, W), where $W = 4\omega/h$. Appendix D gives the residual tolerance and the tighter-tolerance control calculation.

The spatial and temporal comparisons in Sections 5.2 and 5.3 use $\Delta t = 10^{-4}$ and $T = 0.01$; the mesh experiments in Section 5.4 use RK4 with the same step size and final time. In each comparison, the advanced state is expressed in terms of the physical fields u and v for error evaluation. This provides a common basis for assessing the numerical evolution across formulations with different state variables. With the spatial operators and boundary treatment fixed, the time-integrator comparisons then allow us to follow how the temporal approximation influences the accumulation of errors in the two coupled fields.

4.3 A conservation-based moving mesh

The spatial variation of a wave is often concentrated in a small part of the computational domain. A uniform grid distributes nodes equally across these regions and the more slowly varying background, while an adaptive mesh can allocate additional resolution where the fields vary more strongly. In this subsection, we first establish a semi-discrete conservation law for Centered-SD and then use its density and flux to construct a self-adaptive moving mesh. The nodes move in x, while the y lattice remains uniform and all transverse layers share the same longitudinal mesh. This arrangement retains the transverse lattice pairings and allows the distribution of resolution in x to evolve with the physical fields.

To establish the conservation law, we introduce the density $\rho_k = 1 - \omega_k/h$ and define the associated flux q_k by

$$\rho_k = 1 - \frac{\omega_k}{h}, \qquad q_k = (u_k + 2a)\rho_k - \partial_x \rho_k - 2a. \tag{79}$$

Substituting $\omega_k = h(1 - \rho_k)$ into the auxiliary equation of Centered-SD gives $\partial_t \rho_k + \partial_x q_k = 0$. We choose the additive constant in q_k so that the flux vanishes on the zero-field background. This conservation relation connects the evolution of the density to its spatial transport and forms the basis of the mesh construction below.

Since all layers share the same x nodes, we average the density and the flux over the stored layers,

$$\bar{\rho} = \frac{1}{N_y}\sum_k \rho_k, \qquad \bar{q} = \frac{1}{N_y}\sum_k q_k, \tag{80}$$

and the averaged quantities satisfy the same conservation relation.

On a periodic interval of length $2L$, the initial nodes equidistribute the cumulative density. Fixing a reference node x_0, we require

$$\int_{x_0}^{x_i(0)} \bar{\rho}\, dx = \frac{i}{N_x}\int_{x_0}^{x_0+2L} \bar{\rho}\, dx, \qquad i = 0, \ldots, N_x, \tag{81}$$

so intervals in regions of larger $\bar{\rho}$ receive more nodes.

During the evolution each node keeps its cumulative-density label, which is equivalent to preserving the mass of every cell $[x_i, x_{i+1}]$. This requires $\bar{\rho}_i \mathcal{V}_i - \bar{q}_i$ to take the same value at every node. We choose that constant as the negative of the nodal mean of $\bar{q}$, and the velocity is

$$\dot{x}_i = \mathcal{V}_i = \frac{\bar{q}_i - \langle \bar{q} \rangle}{\bar{\rho}_i}, \qquad \langle \bar{q} \rangle = \frac{1}{N_x} \sum_i \bar{q}_i. \tag{82}$$

This choice gives zero density-weighted mean nodal velocity and keeps the mapping periodic. The construction requires $\bar{\rho} > 0$, and the ordering of the nodes is preserved while the discrete Jacobian remains positive.

For the mesh comparison, both the fixed and adaptive calculations use mapped three-point differences in the longitudinal direction. Write $\xi_i = \xi_0 + i\Delta\xi$ for the fixed uniform computational grid and $x_i = \xi_i + s_i$ for the physical nodes. Discretizing the chain rule gives

$$\begin{aligned}
(\widetilde{\partial}_x f)_i &= \frac{f_{i+1} - f_{i-1}}{2\Delta\xi\, J_i}, \\
(\widetilde{\partial}_{xx} f)_i &= \frac{f_{i+1} - 2f_i + f_{i-1}}{(\Delta\xi)^2 J_i^2} \\
&\quad - \frac{(J_{i+1} - J_{i-1})(f_{i+1} - f_{i-1})}{4(\Delta\xi)^2 J_i^3}, \\
J_i &= 1 + \frac{s_{i+1} - s_{i-1}}{2\Delta\xi}.
\end{aligned} \tag{83}$$

Here J_i is the discrete Jacobian, and f_i denotes a sampled field. The formulas use centered three-point differences on the computational grid. Setting $s_i = 0$ gives $J_i = 1$ and the uniform-grid realization used in Section 5.4.

Here i labels the moving longitudinal nodes and k labels the transverse layers. All layers share $x_i(t)$ and its velocity $\mathcal{V}_i$. A dot denotes the total time derivative along a moving node, so we advance

$$\begin{aligned}
\dot{u}_{i,k} &= F_{u,i,k} + \mathcal{V}_i (\widetilde{\partial}_x u)_{i,k}, \\
\dot{\omega}_{i,k} &= F_{\omega,i,k} + \mathcal{V}_i (\widetilde{\partial}_x \omega)_{i,k}, \qquad \dot{x}_i = \mathcal{V}_i.
\end{aligned} \tag{84}$$

Here $F_{u,i,k}$ and $F_{\omega,i,k}$ are the Centered-SD rates at fixed physical position, evaluated with the current mapped operators. The common node velocity retains the reconstruction of v across layers. At the lower transverse boundary $k = 0$, each longitudinal node uses

$$\dot{u}_{i,0} = u_t^*(x_i(t), y_0, t) + \mathcal{V}_i u_x^*(x_i(t), y_0, t).$$

Setting $\mathcal{V}_i = 0$ gives the fixed-grid equations.

The resulting scheme couples the physical-field evolution to a mesh determined by the semi-discrete conservation law. The averaged density specifies the initial distribution of nodes, and the associated flux governs their subsequent motion through the preservation of cumulative-density labels. By sharing the longitudinal nodes across all transverse layers, the construction accommodates mesh adaptation within the existing Centered-SD field reconstruction. It thus provides a concrete way to use the analytical structure of the equations to guide spatial resolution, whose effect on the computed wave fields can be assessed at a fixed node count.

5 Numerical experiments

In this section, we examine how the semi-discrete formulations developed above perform in numerical computation. Our focus is on the accuracy of the evolving physical fields and the roles of spatial discretization, time integration, and mesh distribution in determining that accuracy. We also follow the response to independent initial perturbations to examine how disturbances develop through the coupling of the two fields. Section 5.1 introduces the test problems, continuous reference solutions, and error measures. Section 5.2 examines the spatial approximations and compares the physical fields computed by Endpoint-SD, Centered-SD, and direct

finite differences under common RK4 time integration. With Centered-SD, we then study the error histories of three time integrators in Section 5.3 and the effect of conservation-based mesh adaptation in Section 5.4. Section 5.5 examines the evolution of independent initial perturbations within each spatial formulation.

5.1 Test problems and error measures

We use four initial data sets and their continuous reference solutions, comprising one soliton, two solitons, a periodic Gaussian profile and a sinusoidal profile. All tests use $a = 2$ and $\lambda = -2$. The soliton references follow (6)–(7), with $\rho_i = 1$ and zero phase shifts. Table 1 gives the spectral pairs (p_i, q_i), evaluation windows and computational intervals.

Table 1: Spectral parameters and regions for the soliton tests.

Case	Spectral pairs (p_i, q_i)	Region	Periodic x interval
One-soliton	(1, 2)	R = 5	[-10, 10)
Two-soliton	(1, 2), (4, -3)	R = 30	[-40, 40)

Writing the periodic x interval as $[-L, L)$, we use the same interval in y and place the computational boundaries well outside the region $[-R, R]^2$ containing the principal wave profiles.

As shown in Figure 1, the one-soliton case consists of a single oblique wave band with smooth negative troughs in both u and v.

The two-soliton case (Figure 2) adds the spectral pair $(4, -3)$. Its u field has a negative trough and a shallower positive ridge, while both wave bands are negative in v. The section $y = -20$ displays both soliton profiles.

For the Gaussian and sinusoidal tests, take the separable initial fields

$$\begin{aligned} u_0(x, y) &= A \sin\left[\frac{\pi(y+2)}{4}\right] F_0(x), \\ v_0(x, y) &= 4 - \frac{3\pi A}{4} \cos\left[\frac{\pi(y+2)}{4}\right] F_0(x). \end{aligned} \tag{85}$$

The Gaussian test uses $A = 0.05$ with the periodic localized profile $F_0(x) = \sum_{j\in\mathbb{Z}} \exp[-((x + 192j)/6)^2]$, while the sinusoidal test uses $A = 0.03$ with $F_0(x) = \cos(\pi x/12)$. Both problems are computed on the periodic interval $x \in [-96, 96)$ over the transverse strip $y \in [-2, 2]$ with homogeneous lower boundary data $u = 0$.

These initial fields follow from the heat-equation and Laplace-transformation construction in Appendix E. Taking $s(y) = A \sin[\pi(y+2)/4]/2$ gives $u_0 = 2sF_0$ and $v_0 = 4 - 6s_yF_0$, hence $v_0 = 4 - 3u_{0y}$. Both profiles have explicit continuous references on $0 \le t \le 0.01$. Appendix E establishes the initial matching, nonzero reconstruction denominators and reference-evaluation bounds.

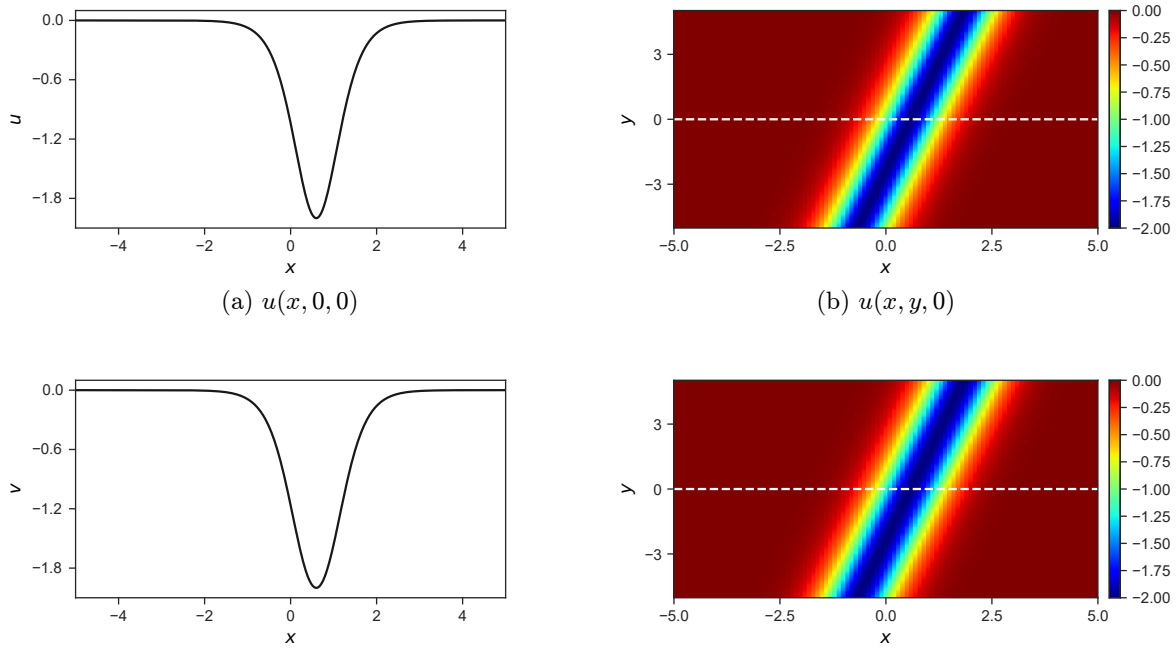
Figure 3 shows the localized Gaussian profile and its transverse variation. We choose $y = 1$ to display a positive hump in u and the corresponding elevation of v above its background value $v = 4$. The heatmaps show how these sections extend across the strip, with the departure of v from its background changing sign across $y = 0$.

The sinusoidal test (Figure 4) replaces the localized pulse in x by a periodic cosine wave, so that u alternates in sign along x while the departure of v from its background $v = 4$ reverses sign across $y = 0$. We choose $y = 1$ to show the periodic wave profiles in both physical fields.

Each numerical method is initialized by sampling the continuous reference fields at its native grid locations. Recall that both semi-discrete formulations place u at transverse nodes and v at edge midpoints, whereas FD places both fields at the nodes. Errors are evaluated directly at these native sites inside the specified interior region ($|x| \le R$ and $|y| < R$ for the soliton tests, and all x nodes with $|y| \le 1.8$ for the Gaussian and sinusoidal tests). At the final time $T = 0.01$, we record the maximum absolute errors

$$E_u(T) = \max_{\mathcal{G}_u} |u_h(T) - u^*(T)|, \qquad E_v(T) = \max_{\mathcal{G}_v} |v_h(T) - v^*(T)|, \tag{86}$$

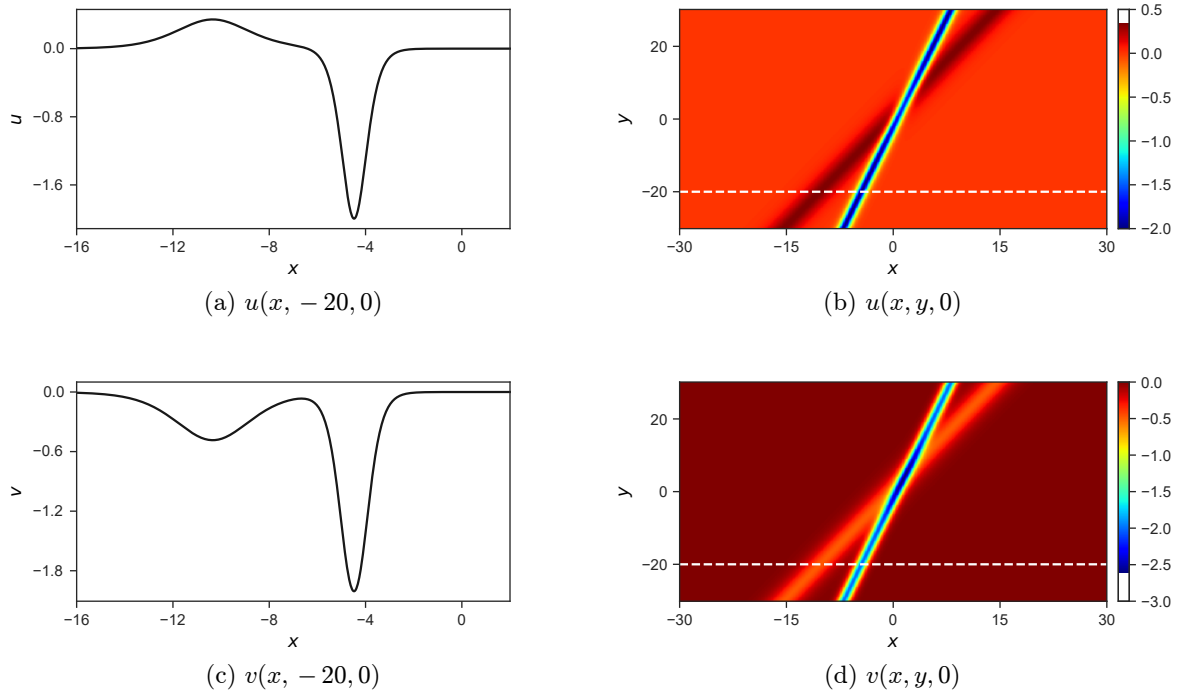
Here, $\mathcal{G}_u$ and $\mathcal{G}_v$ are the native field sites within the evaluation region. The initial derivative errors E_{u_t} and E_{v_t} are the maximum absolute differences between the spatially computed and exact continuous time derivatives at $t = 0$. They measure the spatial approximation before time integration. Bold table entries identify the smallest error for each case and field.



(a) $u(x, 0, 0)$ (b) $u(x, y, 0)$

(c) $v(x, 0, 0)$ (d) $v(x, y, 0)$

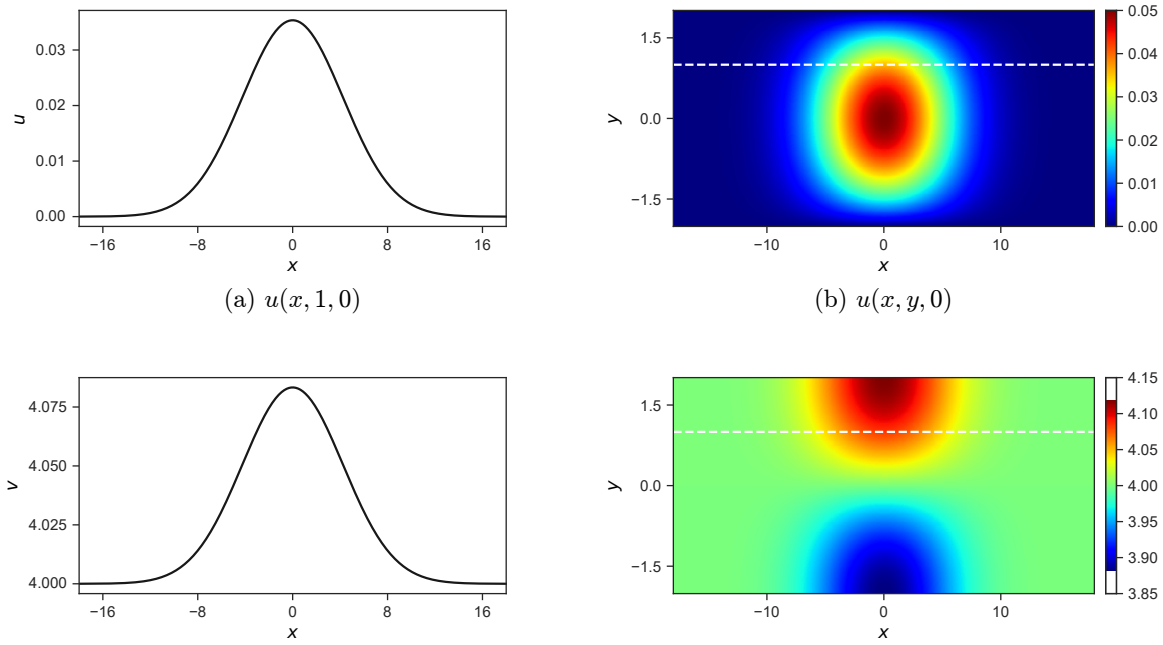
Fig. 1: One-soliton case. Exact initial physical fields. Left: sections at $y = 0$; right: physical-field heatmaps, with the central section marked by a white dashed line. The upper and lower rows show u and v, respectively.



(a) $u(x, -20, 0)$ (b) $u(x, y, 0)$

(c) $v(x, -20, 0)$ (d) $v(x, y, 0)$

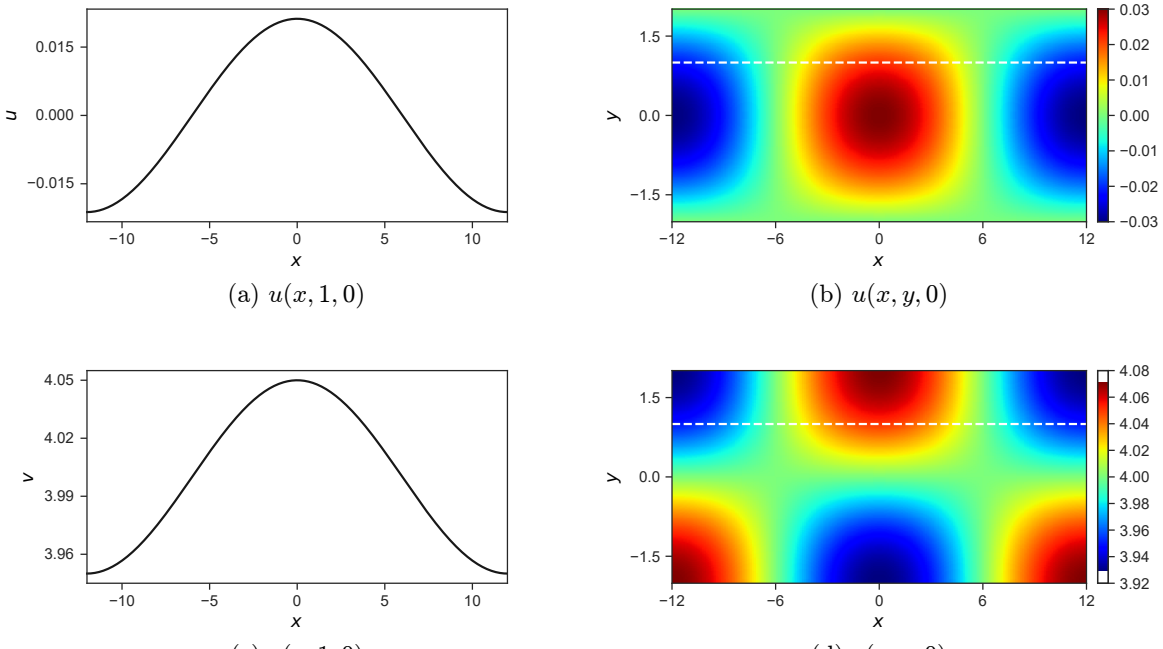
Fig. 2: Two-soliton case. Exact initial physical fields. Left: sections at $y = -20$; right: physical-field heatmaps, with the selected section marked by a white dashed line. The upper and lower rows show u and v, respectively. The section panels focus on $x \in [-16, 2]$; the heatmaps retain the full $[-30, 30]$ window.



(a) $u(x, 1, 0)$ (b) $u(x, y, 0)$

(c) $v(x, 1, 0)$ (d) $v(x, y, 0)$

Fig. 3: Gaussian case. Exact initial physical fields. Left: sections at $y = 1$; right: physical-field heatmaps, with the selected section marked by a white dashed line. The upper and lower rows show u and v, respectively. The displayed x window is $[-18, 18]$; the error tables use the full periodic interval $[-96, 96)$.



(a) $u(x, 1, 0)$ (b) $u(x, y, 0)$

(c) $v(x, 1, 0)$ (d) $v(x, y, 0)$

Fig. 4: Sinusoidal case. Exact initial physical fields. Left: sections at $y = 1$; right: physical-field heatmaps, with the selected section marked by a white dashed line. The upper and lower rows show u and v, respectively. The displayed x window is $[-12, 12]$; the error tables use the full periodic interval $[-96, 96)$.

5.2 Comparison of spatial formulations

In this subsection, we examine how the spatial formulation affects the accuracy of the two physical fields, following the initial spatial approximation through to the computed wave profiles. The analysis in Section 3.4 identifies different truncation errors for Endpoint-SD and Centered-SD, and we use the four test problems to investigate how these differences are reflected in numerical evolution. Direct finite differences applied to the continuous DLW system provide a baseline. All three formulations use $h = \Delta x = 0.1$ and advance from the exact initial data to $T = 0.01$ using classical RK4 with $\Delta t = 10^{-4}$. We first examine the initial time-derivative errors in Table 2, then consider the evolved physical-field errors and their spatial distributions.

Table 2: Initial time-derivative errors at h = Δx = 0.1.

Case	Endpoint-SD		Centered-SD		FD	
	E_{u_t}	E_{v_t}	E_{u_t}	E_{v_t}	E_{u_t}	E_{v_t}
One-soliton	$\mathbf{2.479 \times 10^{-3}}$	5.492×10^{-1}	1.656×10^{-2}	$\mathbf{8.443 \times 10^{-3}}$	4.957×10^{-3}	1.394×10^{-2}
Two-soliton	$\mathbf{2.142 \times 10^{-3}}$	7.010×10^{-1}	1.295×10^{-2}	$\mathbf{1.089 \times 10^{-2}}$	4.284×10^{-3}	1.644×10^{-2}
Gaussian	$\mathbf{2.142 \times 10^{-6}}$	6.038×10^{-5}	2.900×10^{-6}	$\mathbf{6.997 \times 10^{-7}}$	4.284×10^{-6}	2.215×10^{-6}
Sinusoidal	$\mathbf{1.586 \times 10^{-6}}$	2.833×10^{-5}	2.003×10^{-6}	$\mathbf{4.715 \times 10^{-7}}$	3.171×10^{-6}	1.639×10^{-6}

At $t = 0$, the two semi-discrete formulations show a clear division between the two components: Endpoint-SD gives the smallest initial u_t defect E_{u_t} in all four cases, whereas Centered-SD gives the smallest v_t defect E_{v_t}. This pattern is consistent with Proposition 3.4. In the first equation, both formulations are second-order consistent, and the difference in E_{u_t} comes from the $O(h^2)$ auxiliary flux term retained by Centered-SD. In the second equation, however, the one-sided pairing of Endpoint-SD leaves a first-order $O(h)$ truncation term, which makes its initial v_t error significantly larger than those of Centered-SD and FD.

Table 3 follows these initial states to $T = 0.01$ and reports the resulting errors in the physical fields u and v. The cross-section panels compare the continuous reference profiles with the numerical values at $y = 0$ for the one-soliton case, $y = -20$ for the two-soliton case, and $y = 1$ for the Gaussian and sinusoidal cases. A 12-point local polynomial interpolation in y is used when the selected section lies between native lattice lines. The heatmaps display the spatial distribution of the absolute errors across the evaluation region.

Table 3: Physical-field errors at T = 0.01 with RK4, Δt = 10^{-4} and h = Δx = 0.1.

Case	Endpoint-SD		Centered-SD		FD	
	E_u	E_v	E_u	E_v	E_u	E_v
One-soliton	4.702×10^{-4}	5.547×10^{-3}	1.677×10^{-4}	$\mathbf{7.734 \times 10^{-5}}$	$\mathbf{5.568 \times 10^{-5}}$	1.428×10^{-4}
Two-soliton	4.942×10^{-4}	7.076×10^{-3}	1.792×10^{-4}	$\mathbf{1.009 \times 10^{-4}}$	$\mathbf{1.149 \times 10^{-4}}$	1.694×10^{-4}
Gaussian	$\mathbf{2.147 \times 10^{-8}}$	6.036×10^{-7}	2.900×10^{-8}	$\mathbf{6.967 \times 10^{-9}}$	4.280×10^{-8}	2.206×10^{-8}
Sinusoidal	$\mathbf{1.588 \times 10^{-8}}$	2.833×10^{-7}	2.003×10^{-8}	$\mathbf{4.706 \times 10^{-9}}$	3.170×10^{-8}	1.637×10^{-8}

Looking across Table 3, we see that Centered-SD maintains the smallest v error in all four tests, while the ordering of the u error depends on the test problem.

One-soliton and two-soliton cases

Figure 5(a,c) shows the one-soliton profiles along $y = 0$. Both fields form a localized depression, with a single trough and a return toward the background on either side. The numerical symbols from all three formulations follow the continuous reference closely, resolving both the trough and its flanks. Panels (b,d) make their differences visible through the absolute errors along the same section. These errors have several local peaks around the wave, with a particularly clear separation between Endpoint-SD and the other two methods in v.

Comparing the two semi-discrete schemes, we notice that Centered-SD not only reduces the terminal v error relative to Endpoint-SD by a factor of 72 (from 5.547×10^{-3} to 7.734×10^{-5}), but also achieves a smaller terminal u error by a factor of 2.8 (reducing E_u from 4.702×10^{-4} to 1.677×10^{-4}), reversing their initial ordering in Table 2. A simple explanation for this reversal lies in the coupled evolution of the two fields: because the first DLW equation contains the dispersive term v_{xx}, the much larger $O(h)$ error incurred by Endpoint-SD in v feeds back into the time derivative of u as the wave evolves, whereas the symmetric staggered arrangement of Centered-SD keeps both components at second order.

The heatmaps in Figure 5(e–j) extend this section comparison to the two-dimensional region. The dashed horizontal line marks $y = 0$, linking the upper panels to the maps below. Bands of larger error follow the inclined wave structure, with narrow low-error strips separating local maxima. The common color scale within each field makes the smaller v errors of Centered-SD and FD visible alongside the broader, brighter Endpoint-SD error band.

We choose $y = -20$ in Figure 6(a,c), as in Figure 2, to resolve the two soliton profiles in the evolved fields. The u section contains a broad positive crest followed by a narrower negative trough, while v has two distinct negative troughs. The numerical values reproduce these features closely. Panels (b,d) show how the section errors vary across the two waves.

Relative to Endpoint-SD, Centered-SD reduces the terminal u error from 4.942×10^{-4} to 1.792×10^{-4} and the v error from 7.076×10^{-3} to 1.009×10^{-4}. The improvement factors — about 2.8 in u and 70 in v — are nearly identical to those observed in the single-soliton case, and Centered-SD also improves upon the FD baseline in the second field by a factor of about 1.7.

The heatmaps in Figure 6(e–j) show how this error pattern extends across the two-dimensional wave configuration. The upper panels resolve the two waves along $y = -20$, while the maps include their interaction region. In v, the dominant oblique band is substantially brighter for Endpoint-SD; the Centered-SD and FD maps retain a narrower band of larger error. These maps locate the errors reported in Table 3 and show their variation away from the selected section.

Gaussian and sinusoidal cases

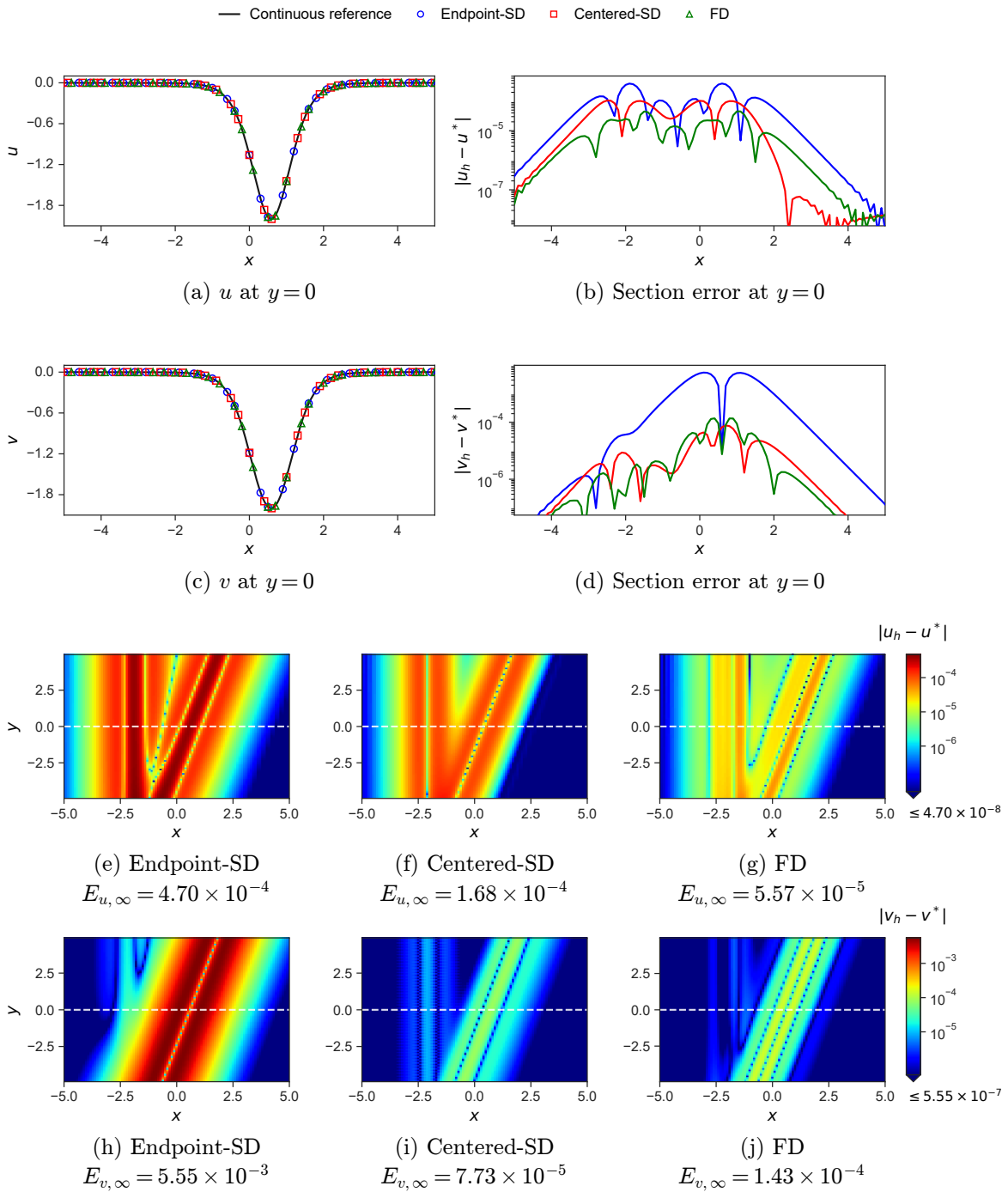
In the smooth, small-amplitude Gaussian (Figure 7) and sinusoidal (Figure 8) tests, the field gradients are much milder and the terminal error ordering at $T = 0.01$ follows the initial derivative defects of Table 2. Endpoint-SD gives the smallest u error, while Centered-SD lies between Endpoint-SD and FD in u and produces the smallest v error in both tests. Specifically, Centered-SD reduces E_v to 6.967×10^{-9} in the Gaussian case (about 3.2 times smaller than FD and 87 times smaller than Endpoint-SD) and to 4.706×10^{-9} in the sinusoidal case (3.5 times smaller than FD and 60 times smaller than Endpoint-SD). For both cases, we use $y = 1$ to display the spatial variation and numerical error of both physical fields.

Figure 7(a,c) shows the evolved Gaussian profiles along $y = 1$, using the same section as Figure 3. Both physical fields retain a smooth localized hump relative to their backgrounds, and the numerical symbols closely follow the reference curves. Panels (b,d) resolve the section errors across these humps, with the smallest maxima attained by Endpoint-SD in u and Centered-SD in v. The heatmaps show how the errors vary across the strip and locate their larger values away from the symmetry line.

Figure 8(a,c) shows the sinusoidal profiles along $y = 1$, using the same section as Figure 4. Both fields have a positive crest relative to their respective backgrounds near the center and lower values toward the ends of the displayed interval. The numerical symbols closely follow the reference curves, while panels (b,d) resolve the errors in each physical field. The section error maxima are smallest for Endpoint-SD in u and Centered-SD in v. The heatmaps extend this comparison across the strip, where the errors form repeated bands and lobes.

The four tests distinguish the accuracy of the initial time derivatives from that of the fields after evolution. Endpoint-SD retains its initial advantage in the first component on the Gaussian and sinusoidal backgrounds, while Centered-SD gives smaller terminal errors in both fields in the soliton tests. The initial approximation of one component must therefore be considered together with the evolution of the coupled fields.

The consistency analysis identifies the truncation terms introduced by each lattice arrangement. During evolution, the coupling provides a route for these errors to affect the other component; in particular, the v_{xx} term links the approximation of v to the computed u field. The change in error ordering in the soliton tests is consistent with this coupling. Initial derivative errors and evolved field errors thus provide complementary measures of the spatial approximation.



(a) u at $y = 0$ (b) Section error at $y = 0$

(c) v at $y = 0$ (d) Section error at $y = 0$

(e) Endpoint-SD
$E_{u,\infty} = 4.70 \times 10^{-4}$

(f) Centered-SD
$E_{u,\infty} = 1.68 \times 10^{-4}$

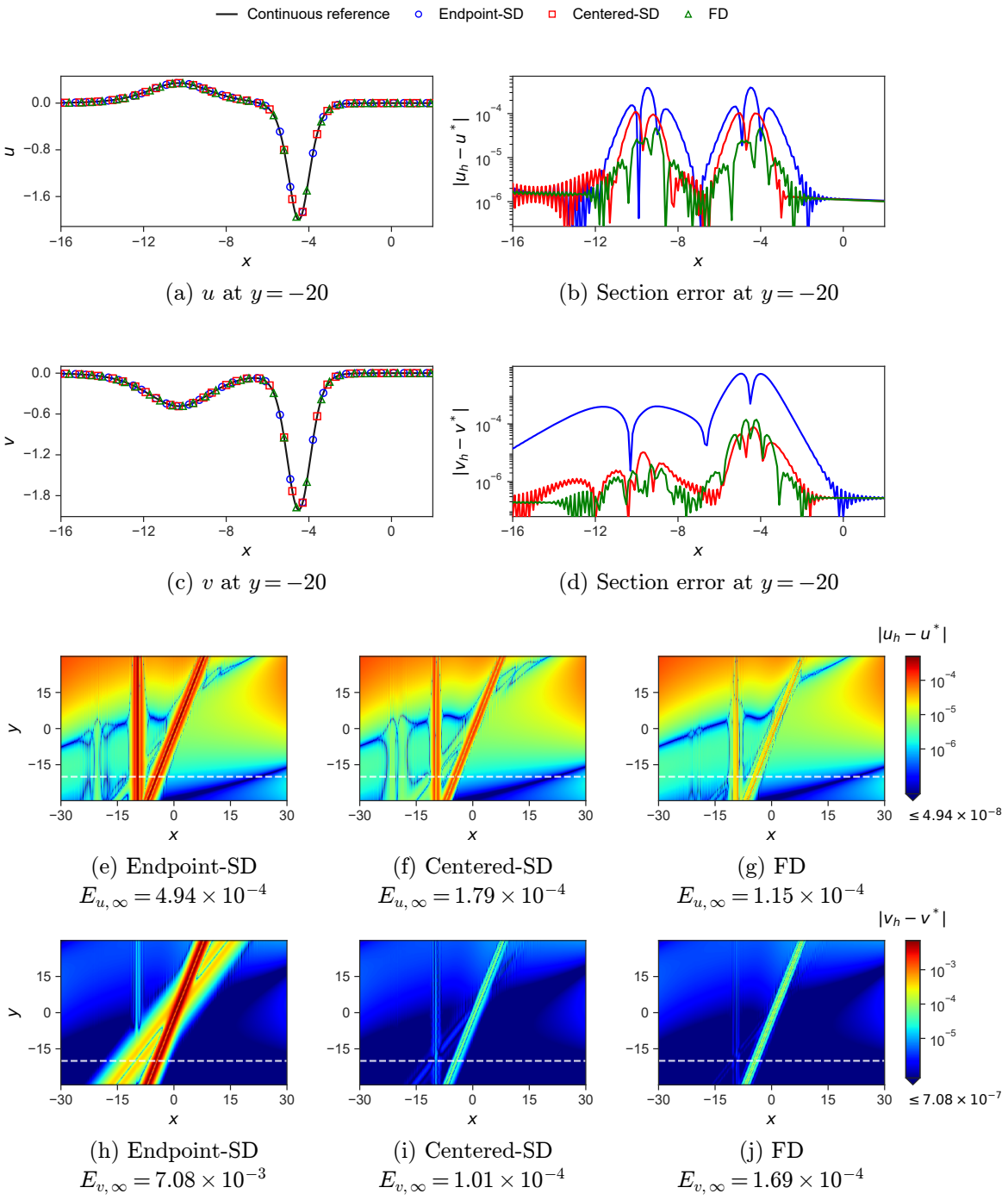
(g) FD
$E_{u,\infty} = 5.57 \times 10^{-5}$

(h) Endpoint-SD
$E_{v,\infty} = 5.55 \times 10^{-3}$

(i) Centered-SD
$E_{v,\infty} = 7.73 \times 10^{-5}$

(j) FD
$E_{v,\infty} = 1.43 \times 10^{-4}$

Fig. 5: One-soliton case at $T = 0.01$. Panels (a,c) compare the continuous $y = 0$ profiles with numerical symbols; panels (b,d) show the absolute errors along that section. The lower rows show the u and v error maps, with columns for Endpoint-SD, Centered-SD and FD. Each row uses one logarithmic color scale; values at or below its lower limit, including zero, share the lowest (dark-blue) color, as indicated by the $\leq$ label on the colorbar. Panel maxima refer to the full table region.



(a) u at $y = -20$ (b) Section error at $y = -20$

(c) v at $y = -20$ (d) Section error at $y = -20$

(e) Endpoint-SD
$E_{u,\infty} = 4.94 \times 10^{-4}$

(f) Centered-SD
$E_{u,\infty} = 1.79 \times 10^{-4}$

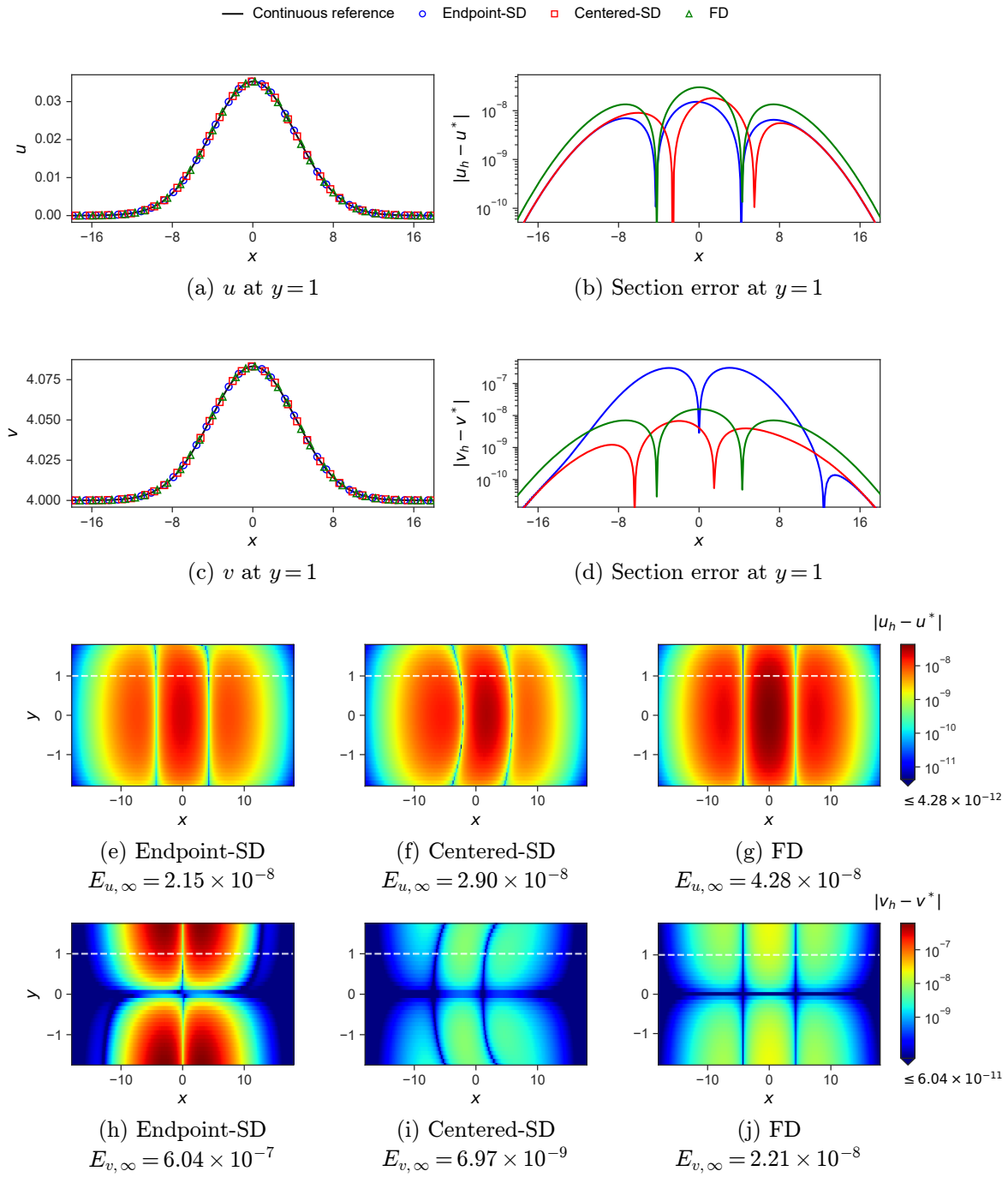
(g) FD
$E_{u,\infty} = 1.15 \times 10^{-4}$

(h) Endpoint-SD
$E_{v,\infty} = 7.08 \times 10^{-3}$

(i) Centered-SD
$E_{v,\infty} = 1.01 \times 10^{-4}$

(j) FD
$E_{v,\infty} = 1.69 \times 10^{-4}$

Fig. 6: Two-soliton case at $T = 0.01$. Panels (a,c) compare the continuous $y = -20$ profiles with numerical symbols; panels (b,d) show the absolute errors along that section. The lower rows show the u and v error maps, with columns for Endpoint-SD, Centered-SD and FD. Each row uses one logarithmic color scale; values at or below its lower limit, including zero, share the lowest (dark-blue) color, as indicated by the $\leq$ label on the colorbar. Panel maxima refer to the full table region. The sections focus on $x \in [-16, 2]$, while the maps retain $[-30, 30]$.



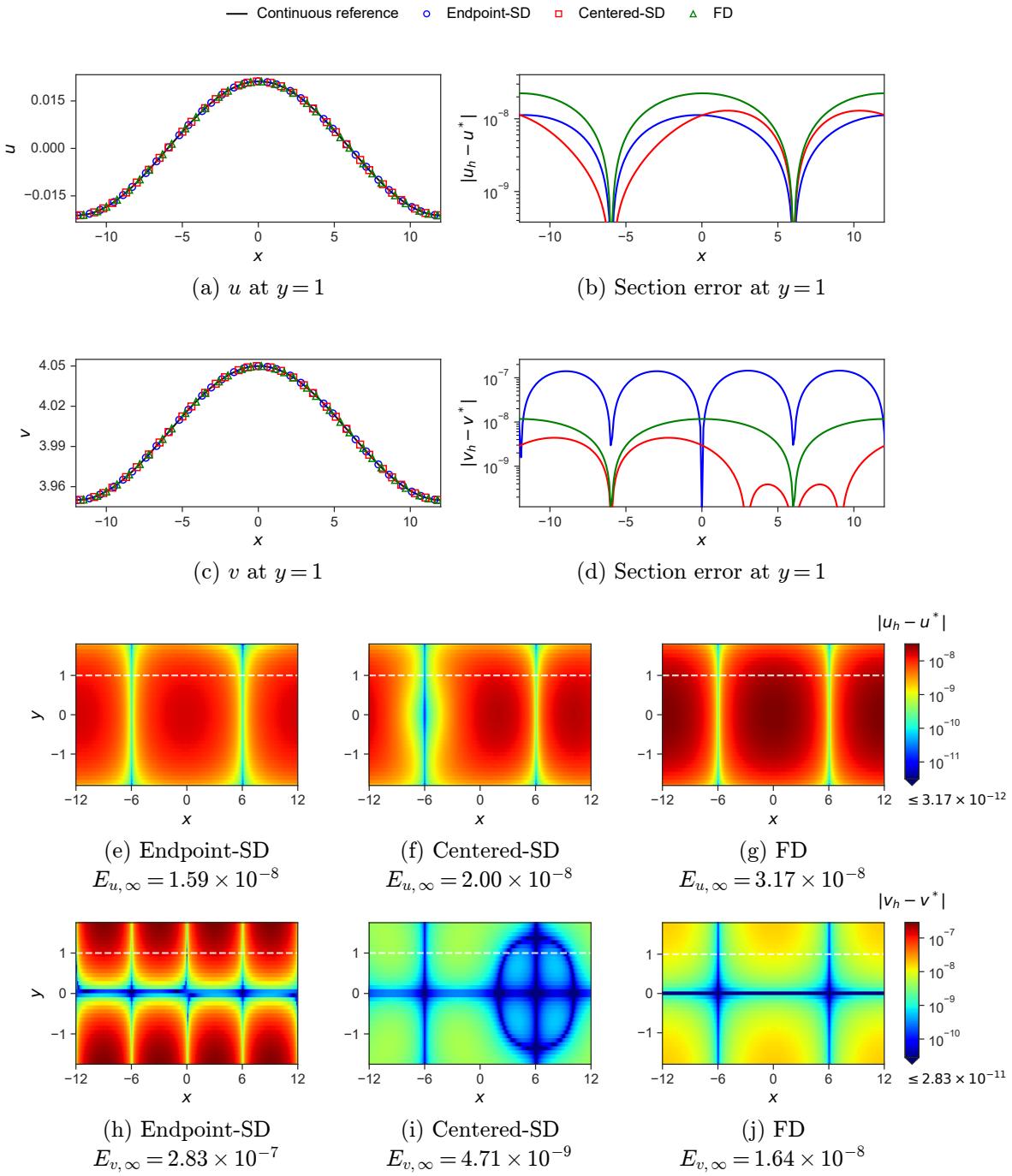
(a) u at $y = 1$ (b) Section error at $y = 1$

(c) v at $y = 1$ (d) Section error at $y = 1$

(e) Endpoint-SD $E_{u,\infty} = 2.15 \times 10^{-8}$ (f) Centered-SD $E_{u,\infty} = 2.90 \times 10^{-8}$ (g) FD $E_{u,\infty} = 4.28 \times 10^{-8}$

(h) Endpoint-SD $E_{v,\infty} = 6.04 \times 10^{-7}$ (i) Centered-SD $E_{v,\infty} = 6.97 \times 10^{-9}$ (j) FD $E_{v,\infty} = 2.21 \times 10^{-8}$

Fig. 7: Gaussian case at $T = 0.01$. Panels (a,c) compare the continuous $y = 1$ profiles with numerical symbols; panels (b,d) show the absolute errors along that section. The lower rows show the u and v error maps, with columns for Endpoint-SD, Centered-SD and FD. Each row uses one logarithmic color scale; values at or below its lower limit, including zero, share the lowest (dark-blue) color, as indicated by the $\leq$ label on the colorbar. Panel maxima refer to the full table region. The display window is $[-18, 18]$; the table uses the full periodic interval $[-96, 96)$.



(a) u at $y = 1$ (b) Section error at $y = 1$

(c) v at $y = 1$ (d) Section error at $y = 1$

(e) Endpoint-SD $E_{u,\infty} = 1.59 \times 10^{-8}$ (f) Centered-SD $E_{u,\infty} = 2.00 \times 10^{-8}$ (g) FD $E_{u,\infty} = 3.17 \times 10^{-8}$

(h) Endpoint-SD $E_{v,\infty} = 2.83 \times 10^{-7}$ (i) Centered-SD $E_{v,\infty} = 4.71 \times 10^{-9}$ (j) FD $E_{v,\infty} = 1.64 \times 10^{-8}$

Fig. 8: Sinusoidal case at $T = 0.01$. Panels (a,c) compare the continuous $y = 1$ profiles with numerical symbols; panels (b,d) show the absolute errors along that section. The lower rows show the u and v error maps, with columns for Endpoint-SD, Centered-SD and FD. Each row uses one logarithmic color scale; values at or below its lower limit, including zero, share the lowest (dark-blue) color, as indicated by the $\leq$ label on the colorbar. Panel maxima refer to the full table region. The display window is $[-12, 12]$; the table uses the full periodic interval $[-96, 96)$.

5.3 Comparison of time integrators

The preceding comparison follows spatial approximation errors through the coupled evolution using RK4. We now keep Centered-SD fixed and vary the time integrator to examine how temporal approximation changes this behavior. Forward Euler, classical RK4, and Crank–Nicolson use the same exact initial fields, spatial operators and boundary data on the uniform grid with $h = \Delta x = 0.1$, and advance to $T = 0.01$ with $\Delta t = 10^{-4}$. We compare both the terminal errors and their time histories to see how the additional temporal error affects each physical component.

Table 4: Physical-field errors at T = 0.01 for Centered-SD with forward Euler, RK4 and Crank–Nicolson at h = Δx = 0.1 and Δt = 10^{-4}. Values are rounded to four significant digits; some RK4 and Crank–Nicolson entries therefore coincide. Additional digits, signed differences and the smaller recorded error are given in Appendix D, Table 8. Bold values indicate the smallest error among the three integrators, determined before rounding.

Case	Forward Euler		RK4		Crank–Nicolson	
	E_u	E_v	E_u	E_v	E_u	E_v
One-soliton	$\mathbf{1.664 \times 10^{-4}}$	8.026×10^{-5}	1.677×10^{-4}	$\mathbf{7.734 \times 10^{-5}}$	1.677×10^{-4}	7.734×10^{-5}
Two-soliton	$\mathbf{1.420 \times 10^{-4}}$	1.090×10^{-4}	1.792×10^{-4}	$\mathbf{1.009 \times 10^{-4}}$	1.805×10^{-4}	1.009×10^{-4}
Gaussian	$\mathbf{5.107 \times 10^{-9}}$	5.680×10^{-8}	2.900×10^{-8}	$\mathbf{6.967 \times 10^{-9}}$	2.900×10^{-8}	6.969×10^{-9}
Sinusoidal	$\mathbf{1.243 \times 10^{-8}}$	4.213×10^{-8}	2.003×10^{-8}	4.706×10^{-9}	2.003×10^{-8}	$\mathbf{4.705 \times 10^{-9}}$

Table 4 reports the terminal errors. RK4 reproduces the Centered-SD values of Table 3, and Crank–Nicolson differs from them by at most 0.73%. Some entries coincide at the displayed precision; Table 8 gives their signed differences and the step-size checks. Forward Euler gives a smaller u error and a larger v error in all four tests, so the terminal ranking depends on the field component.

To see how the errors accumulate during the computation, we record the maximum absolute error over the native grid sites at every time step $t_n = n\Delta t$,

$$E_u^{(m)}(t_n) = \max_{\mathcal{G}_u} \left| u_h^{(m)}(t_n) - u^*(t_n) \right|, \qquad E_v^{(m)}(t_n) = \max_{\mathcal{G}_v} \left| v_h^{(m)}(t_n) - v^*(t_n) \right|, \tag{87}$$

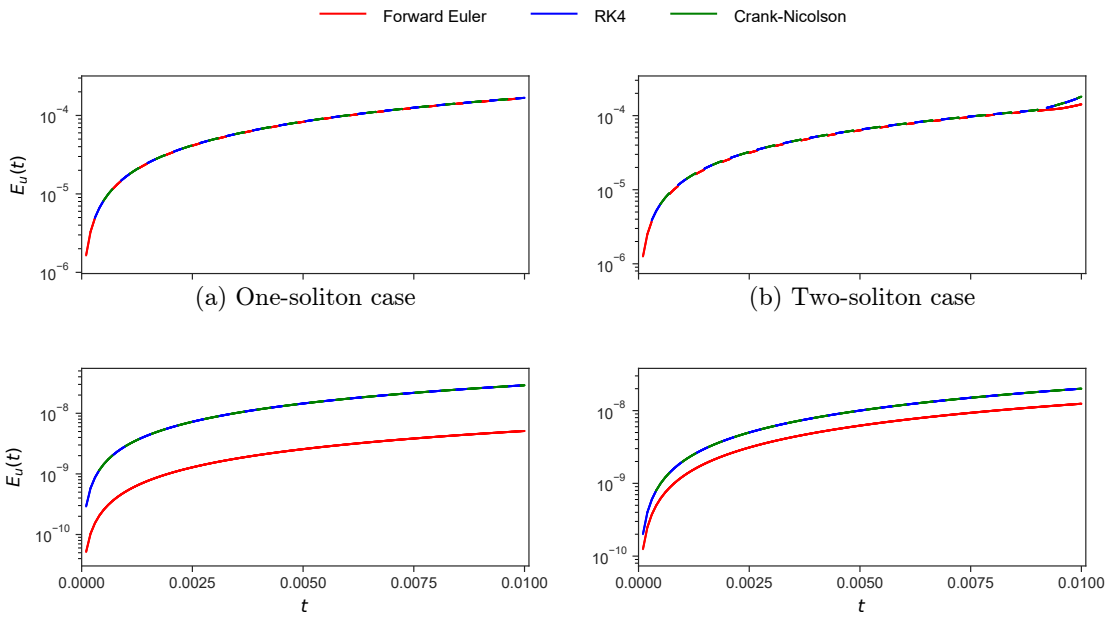
where m labels the time integrator. Because the grid point attaining the maximum can shift as the wave moves, these curves track the evolution of the worst-case error in each field across the domain.

Figure 9 follows the maximum error in u. The one-soliton, Gaussian, and sinusoidal cases show nearly linear accumulation, with $E_u(t)/t$ varying by less than 10%. The two-soliton curve steepens near the end of the interval, with this ratio increasing by 36% over the final tenth of the run. RK4 and Crank–Nicolson nearly coincide, while the Euler curve remains below them.

Figure 10 gives the corresponding v histories. The errors grow smoothly from zero. Euler follows the same general pattern at a higher level, with the clearest separation on the Gaussian and sinusoidal backgrounds.

The opposite shifts in the two field errors suggest that the temporal contribution partly cancels the spatial error in u and reinforces it in v. To examine this interpretation, we halve the Euler time step to $\Delta t = 5\times10^{-5}$. The terminal error pair moves from (1.664×10^{-4}, 8.026×10^{-5}) to (1.671×10^{-4}, 7.880×10^{-5}) in the one-soliton test, and from (1.420×10^{-4}, 1.090×10^{-4}) to (1.560×10^{-4}, 1.048×10^{-4}) in the two-soliton test. In both cases, the errors move toward the RK4 values. This step-size response supports the cancellation and reinforcement interpretation and helps explain the smaller terminal u errors obtained with Euler at the tested step size.

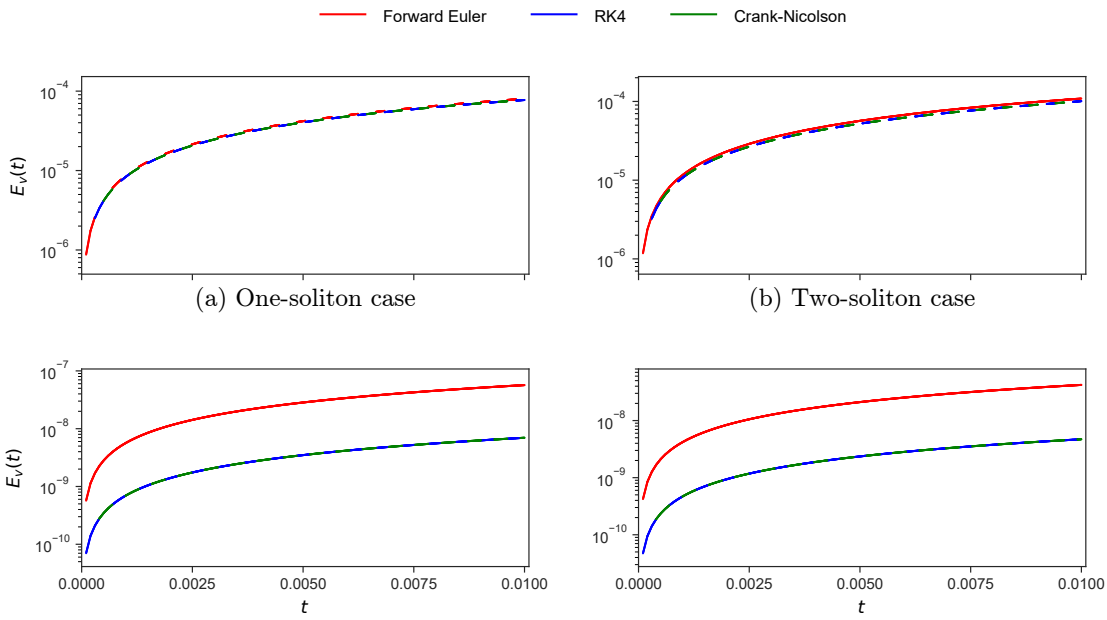
The similar shapes of the error histories suggest that the spatial approximation and wave evolution largely determine the pattern of accumulation, while the time integrator changes its magnitude in each component. Read together with the spatial comparison, these results show why the initial derivative accuracy, the terminal field error and the response to a smaller time step describe different aspects of the same computation.



(a) One-soliton case (b) Two-soliton case

(c) Gaussian case (d) Sinusoidal case

Fig. 9: Development of the maximum absolute error in u for Centered-SD with forward Euler (red), RK4 (blue) and Crank–Nicolson (green): (a) one-soliton, (b) two-soliton, (c) Gaussian and (d) sinusoidal cases. Each value is the maximum over the native field locations in the region of Section 5.1 and is evaluated at every time step; the vertical axis is logarithmic. All calculations use $h = \Delta x = 0.1$ and $\Delta t = 10^{-4}$. The initial error vanishes. Overlapping curves use alternating colors, with solid strokes where they separate.



(a) One-soliton case (b) Two-soliton case

(c) Gaussian case (d) Sinusoidal case

Fig. 10: Development of the maximum absolute error in v, with the same colors as Figure 9. Each value is the maximum over the native midpoints $y_{k-1/2}$ of the second field in the region of Section 5.1. Overlapping curves use alternating colors, with solid strokes where they separate.

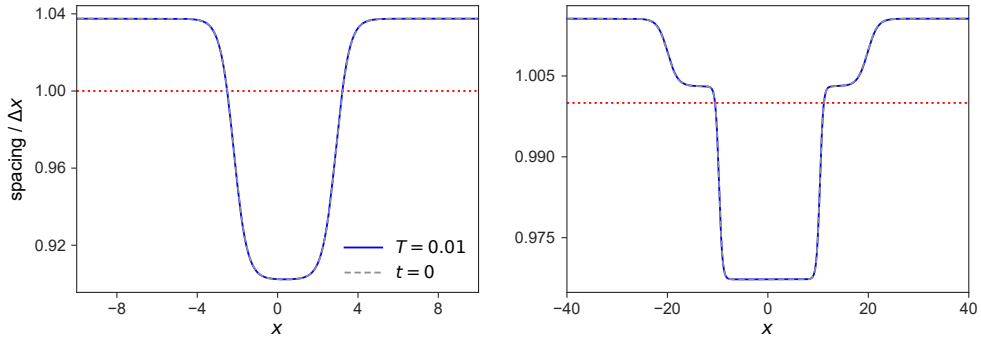
5.4 Fixed and adaptive meshes

In this subsection, we use the semi-discrete conservation law to examine how the distribution of spatial resolution affects the accuracy of Centered-SD. The soliton fields concentrate their variation along localized wave bands, and the conservation density provides a means of placing additional nodes in these regions. We compare uniform and adaptive meshes for the one- and two-soliton problems, following both the node distribution and the resulting physical-field errors. This allows us to examine how a conserved quantity from the semi-discrete system can guide the numerical resolution of its waves.

Both calculations use the same number of shared x nodes, the same uniform transverse lattice with $h = 0.1$, and RK4 with $\Delta t = 10^{-4}$ up to $T = 0.01$. The uniform x spacing is $\Delta x = 0.1$. On both meshes, the x derivatives are evaluated using the mapped three-point centered difference operators of Section 4.3, so that the comparison uses a common spatial approximation while varying the node distribution. The soliton backgrounds provide a positive transverse-averaged conservation density for the equidistribution construction.

In the adaptive runs, the initial x nodes are placed by equidistributing the transverse-averaged density $\bar{\rho}$ computed from the exact initial fields, and the nodes subsequently evolve according to the conservation-based velocity of Section 4.3. The averaged density ranges from 1.000 in the background to 1.150 across the wave core in the one-soliton case, and from 1.000 to 1.050 in the wider domain of the two-soliton case. Figure 11 plots the resulting local grid spacing, normalized by the uniform spacing $\Delta x = 0.1$, at $t = 0$ and $T = 0.01$.

As seen in Figure 11, the equidistribution concentrates nodes in the region of larger density. In the one-soliton test, the local spacing contracts to $0.90\Delta x$ across the wave trough while stretching to $1.04\Delta x$ in the background; in the two-soliton test, it varies more mildly between $0.97\Delta x$ and $1.02\Delta x$. Over the short evolution to $T = 0.01$, the initial and final node distributions are nearly indistinguishable: the maximum node displacement is 9.5×10^{-4} in the one-soliton case and 5.2×10^{-4} in the two-soliton case, and the total discrete mass of $\bar{\rho}$ is conserved to within 10^{-7} relative error. The spatial resolution pattern is thus established by the initial equidistribution and adjusted only slightly by the node motion during the run.



(a) One-soliton case (b) Two-soliton case

Fig. 11: Local x spacing normalized by the uniform spacing $\Delta x = 0.1$, plotted at the interval midpoints of (a) the one-soliton and (b) the two-soliton case. Grey dashed: initial adaptive mesh; blue: $T = 0.01$; red dotted: uniform spacing.

Table 5 compares the terminal errors at $T = 0.01$ on the uniform and adaptive meshes, measuring the relative improvement via

$$\text{Reduction } (\%) = 100 \left(1 - \frac{E_f^{\text{adaptive}}(T)}{E_f^{\text{uniform}}(T)} \right), \qquad f = u, v. \tag{88}$$

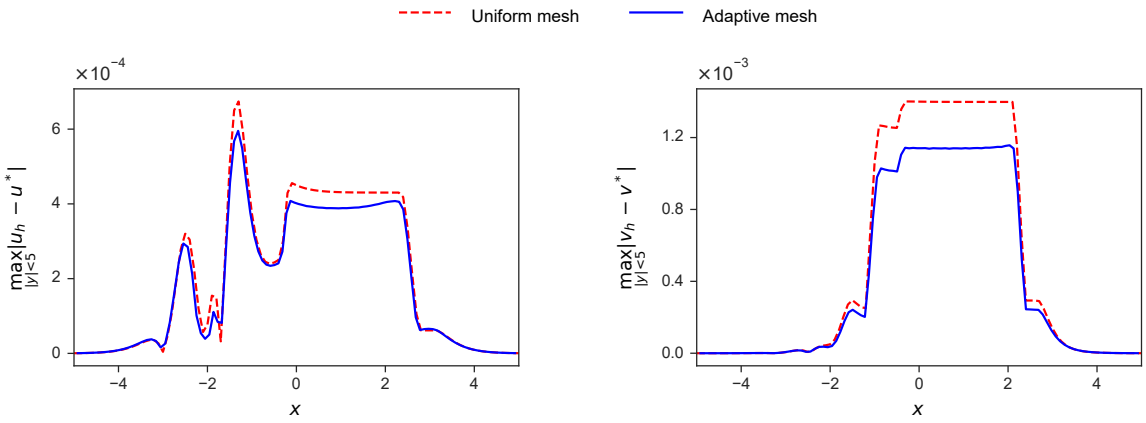
Positive percentages in (88) indicate a reduction in error. The adaptive mesh lowers the errors in both physical fields for both test problems: by 11.7% in u and 17.4% in v for the one-soliton case, and by 6.0% in u and 5.6% in v for the two-soliton case. These improvements are modest and commensurate with the 3%–10% local spacing reductions shown in Figure 11, but they occur consistently across both components without degrading either field. Moreover, a control calculation in which the initially equidistributed nodes are held fixed in time ($\mathcal{V}_i = 0$) yields reductions of 12.8% and 16.3% for the one-soliton case and 5.3% and

Table 5: Physical-field errors at T = 0.01 on the uniform and adaptive meshes, with the relative change defined in the text. Both calculations use Centered-SD with mapped centred x differences, RK4, $\Delta t = 10^{-4}$ and the same number of nodes.

Case	Uniform mesh		Adaptive mesh		Reduction (%)	
	E_u	E_v	E_u	E_v	u	v
One-soliton	6.741×10^{-4}	1.401×10^{-3}	$\mathbf{5.954 \times 10^{-4}}$	$\mathbf{1.157 \times 10^{-3}}$	11.7%	17.4%
Two-soliton	5.533×10^{-4}	1.589×10^{-3}	$\mathbf{5.198 \times 10^{-4}}$	$\mathbf{1.501 \times 10^{-3}}$	6.0%	5.6%

5.8% for the two-soliton case — within 1.1 percentage points of the moving-mesh values in Table 5. Over this short time interval, the accuracy gain therefore comes primarily from the initial density-adapted node placement rather than from dynamic node motion.

To show where this reduction takes place in space, Figure 12 plots the transverse-maximum error profiles $\max_{|y|<5} |u_h - u^*|$ and $\max_{|y|<5} |v_h - v^*|$ as functions of x for the one-soliton case. On the uniform mesh, the numerical error is not distributed evenly across the domain, but is localized along the central wave band $x \in [-3, 3]$. Because the conservation density $\bar{\rho}$ peaks in this same band, the equidistribution shifts nodes from the quiet background directly into the region where the truncation error is generated. As a result, the adaptive mesh lowers the maximum errors and the main error peaks — reducing the main peak in u from 6.74×10^{-4} to 5.95×10^{-4} and lowering the broad plateau in v from 1.40×10^{-3} to 1.16×10^{-3}. We also notice that the relative gain in the one-soliton test is larger for v (17.4%) than for u (11.7%). This is natural in the Centered-SD formulation: the monitor density $\rho_k = 1 - \omega_k/h$ is a shifted rescaling of the auxiliary variable ω_k, which enters the staggered reconstruction $v_k = \Delta_h u_k + \frac{2}{h}(\omega_k + \omega_{k-1})$ and specifically weights the part of v not captured by the backward difference of u.



(a) u (b) v

Fig. 12: One-soliton case at $T = 0.01$. Maximum over the transverse layers of the absolute error at each native x node for the uniform mesh (red) and the adaptive mesh (blue): (a) u and (b) v. Both calculations use the mapped centred operators of Section 4.3, the same number of nodes and RK4 with $\Delta t = 10^{-4}$.

The mesh experiments show that the semi-discrete conservation density provides a useful basis for allocating spatial resolution. Equidistribution concentrates nodes along the soliton wave bands, where the uniform-grid errors are also largest, and reduces the errors in both physical fields at the same node count. The spatial error profiles further show that this improvement extends across the wave region, connecting the change in node placement with the location of the computed errors.

Over the present time interval, the node distribution changes only slightly, and the calculation with a fixed, initially adapted mesh gives similar error reductions. The main benefit therefore comes from the resolution pattern established by the conservation density. These results give the conservation law a concrete role in numerical computation: it guides the placement of nodes in a way that improves the approximation

of the physical fields. The mesh construction thus provides a further connection between the analytical structure of the semi-discrete system and the accurate computation of its waves.

5.5 Evolution of independent physical-field perturbations

The preceding experiments start from prescribed reference wave fields and examine how accurately their evolution is reproduced. A related question concerns the response of the numerical evolution when the initial physical fields are slightly changed. Such changes may arise when initial data are sampled or prescribed, and their influence develops through the coupling between the two fields. A disturbance introduced into u can generate a response in v, while a disturbance in v can also affect the subsequent evolution of u. Understanding this response helps describe how the computed waves depend on their initial physical data.

We examine this dependence on the same four backgrounds by perturbing u_0 and v_0 separately. For each spatial formulation, the perturbed evolution is measured relative to its own unperturbed trajectory. This separates the continuing response in the initially disturbed field from the response generated in the other component, allowing us to follow both processes over time. Let y_L, y_R be the endpoints of the transverse node interval for the chosen test. The two experiments add $\varepsilon\Phi$ to u_0 or v_0, respectively, with

$$\Phi(x,y) = \sin^2\!\left(\frac{\pi(y-y_L)}{y_R-y_L}\right)\cos\!\left(\frac{\pi x}{2}\right), \qquad \varepsilon = 10^{-6}. \tag{89}$$

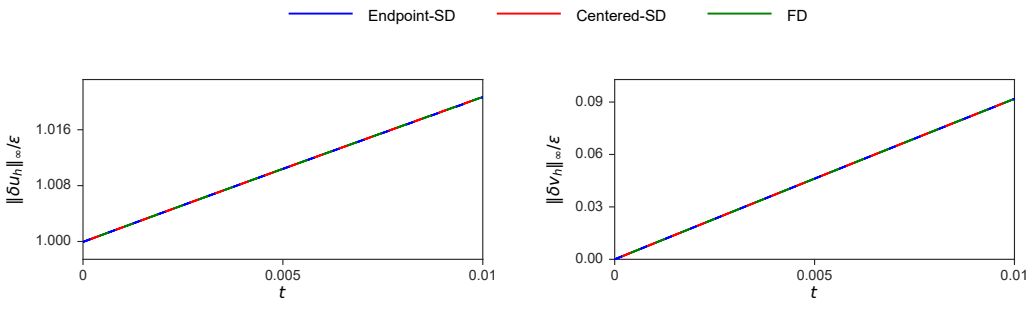
The perturbation is sampled at each field's native sites. The transverse node intervals are $[-9.95, 9.95]$ for the one-soliton case, $[-39.95, 39.95]$ for the two-soliton case, and $[-2, 2]$ for the Gaussian and sinusoidal cases. The envelope and its first derivative vanish at the transverse endpoints. All trajectories use $h = \Delta x = 0.1$, RK4, $\Delta t = 10^{-4}$, and $T = 0.01$, with the prescribed lower u trace and the boundary closures of Section 4.2.

For each spatial formulation, let $(u_h^\varepsilon, v_h^\varepsilon)$ and (u_h^0, v_h^0) be its perturbed and unperturbed numerical trajectories. We record $\delta u_h = u_h^\varepsilon - u_h^0$ and $\delta v_h = v_h^\varepsilon - v_h^0$, and plot their native-node maximum magnitudes divided by ε in the observation regions of Section 5.1. The initial ordinates are the maxima of the imposed perturbation sampled at those same sites. The nonlinear difference equations are advanced together with the corresponding background at every RK4 stage.

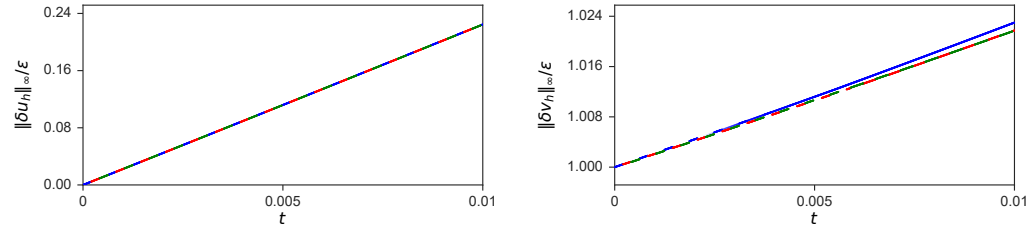
For the one-soliton background, Figure 13 follows both perturbation directions. With u_0 perturbed, the response in u remains near its initial amplitude and grows slightly, while the initially unperturbed v field develops a response from zero. For Endpoint-SD, Centered-SD, and FD, the terminal induced v responses are 0.09182, 0.09186, and 0.09185 times ε, respectively. With v_0 perturbed, each formulation produces a growing u response reaching approximately 0.225ε, alongside a small increase in the initially imposed v disturbance. The two inputs thus expose both the persistence of the original disturbance and the response generated through coupling.

The two-soliton background produces a substantial induced u response when the initial perturbation is applied to v_0 (Figure 14). By $T = 0.01$, this response reaches approximately 0.974ε within each formulation, while the v response remains close to its initial magnitude. For the reverse input, an initial u_0 perturbation produces terminal v responses of 0.10280ε, 0.10278ε, and 0.10277ε for Endpoint-SD, Centered-SD, and FD, respectively. These histories show how a disturbance initially confined to one component develops into a response in both physical fields on the interacting-wave background.

On the Gaussian and sinusoidal backgrounds, the initially perturbed component stays close to its initial magnitude over the computed interval, while a smaller response develops in the other field. Table 6 records the terminal responses separately for each formulation. An initial u_0 disturbance induces a v response of approximately 0.019ε, and a v_0 disturbance induces a u response of approximately 0.049ε. The full time histories are included in the supplementary response plots.

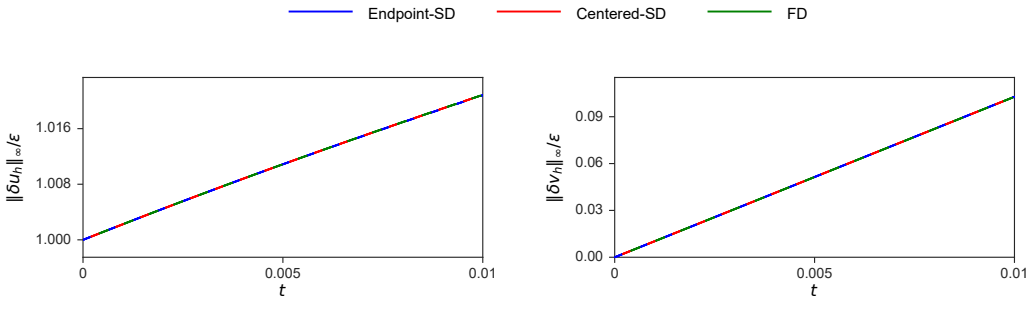


(a) u_0 perturbed: u response (b) u_0 perturbed: v response

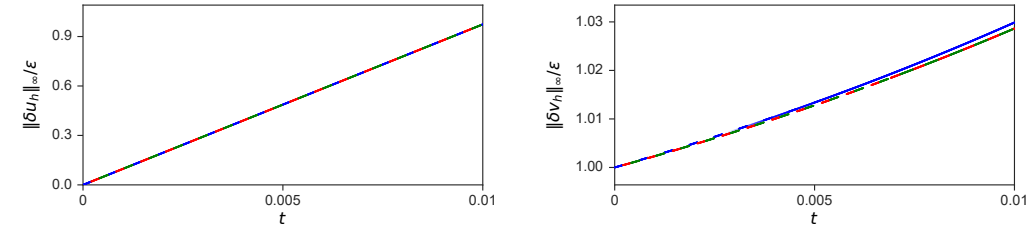


(c) v_0 perturbed: u response (d) v_0 perturbed: v response

Fig. 13: One-soliton perturbation responses. The upper and lower rows perturb u_0 and v_0, respectively; the columns show the normalized u and v responses. Blue: Endpoint-SD; red: Centered-SD; green: FD. Each curve uses its own unperturbed trajectory and native field sites. Where curves overlap, their colors alternate; separated curves are solid. The Endpoint-SD response in panel (d) is slightly larger than the other two responses.



(a) u_0 perturbed: u response (b) u_0 perturbed: v response



(c) v_0 perturbed: u response (d) v_0 perturbed: v response

Fig. 14: Two-soliton perturbation responses. The upper and lower rows perturb u_0 and v_0, respectively; the columns show the normalized u and v responses. Blue: Endpoint-SD; red: Centered-SD; green: FD. Each curve uses its own unperturbed trajectory and native field sites. Where curves overlap, their colors alternate; separated curves are solid. The Endpoint-SD response in panel (d) is slightly larger than the other two responses.

Table 6: Terminal normalized responses on the Gaussian and sinusoidal backgrounds at $T = 0.01$. Each pair gives $(\|\delta u_h\|_\infty/\varepsilon, \|\delta v_h\|_\infty/\varepsilon)$ for the indicated initial perturbation.

Method	Background	u_0 perturbed	v_0 perturbed
Endpoint-SD	Gaussian	(0.99837, 0.01935)	(0.04922, 0.99684)
	Sinusoidal	(0.99841, 0.01933)	(0.04922, 0.99687)
Centered-SD	Gaussian	(0.99837, 0.01935)	(0.04922, 0.99683)
	Sinusoidal	(0.99841, 0.01933)	(0.04922, 0.99687)
FD	Gaussian	(0.99837, 0.01935)	(0.04921, 0.99837)
	Sinusoidal	(0.99841, 0.01933)	(0.04922, 0.99841)

These experiments show how a small change in one initial physical field develops through the coupled numerical evolution. Within each formulation, the initially perturbed component remains close to its original disturbance magnitude over the computed interval, while the other field acquires a response whose magnitude depends on the background and the perturbed component. The substantial induced u response in the two-soliton case is consistent with the transverse integration in the first field equation, through which the influence of a v perturbation can accumulate across the domain. Following the two inputs separately therefore helps us understand both the persistence of an initial disturbance and the response generated by the coupling. Alongside the preceding accuracy tests, these histories describe how the computed waves respond to changes in their physical initial data.

6 Conclusion and discussion

We have developed two related integrable semi-discretizations of the $(2+1)$-dimensional DLW system within a common bilinear framework. Starting from the Hirota–Miwa equation, the construction yields endpoint and centered formulations with explicit Gram determinant solutions, a common positive parameter class, and compatible Darboux–Lax representations. Their physical-field equations make the consequences of lattice placement and reconstruction explicit. Endpoint-SD is second-order consistent in the first equation and first-order consistent in the second, whereas Centered-SD is second-order consistent in both on the specified smooth reconstructed branches. Matching the centered Gram solutions to the continuous family further establishes second-order approximation of both exact physical fields under the stated regularity and positivity conditions.

The numerical experiments examine how these analytical properties are reflected in the computed waves. Endpoint-SD gives the smallest initial u_t error in all four tests and retains its advantage in the terminal u error on the Gaussian and sinusoidal backgrounds, while Centered-SD gives smaller terminal errors in both fields in the soliton tests. Time-step controls support component-dependent cancellation and reinforcement of spatial and temporal errors. The conservation-based mesh improves both field errors at a fixed node count in the soliton tests, with most of the short-time gain arising from the initial node distribution. Independent initial perturbations reveal the responses generated through coupling between the fields. Together, these experiments show how wave profile, field component, and numerical realization affect the accuracy and response of the discrete evolution.

Taken together, the analytical and numerical results clarify the connection between integrable lattice structure and physical-wave approximation. The shared determinant and Lax constructions allow the two formulations to be compared within a common setting, while their reconstructed equations expose differences in consistency and coupling. Exact solutions then connect the continuum analysis to measurable field errors. This comparison shows why the assessment of a lattice benefits from following its consequences through several stages, from the bilinear pairings and physical-field reconstruction to the approximation of exact waves and their numerical evolution. The observed dependence on background and component also gives context to the formal consistency orders.

Beyond the DLW system, this study suggests a way to organize the analysis and computation of integrable wave equations around the physical fields they describe. Related discrete constructions can be examined through their reconstructed equations, matched exact solutions, and componentwise numerical errors, while

available conservation laws can inform the allocation of resolution. Applying this approach to other multi-component systems will require establishing the relevant reconstruction, regularity, and continuum matching conditions for each model. Such studies could help identify which features of an integrable construction determine approximation quality and how those features can guide the design of numerical methods for nonlinear waves.

Appendix A. Determinant identities and alternative solution representations

Every field used in the main text is a logarithmic derivative of a τ function, so its evaluation reduces to an identity for a determinant and for the derivatives of that determinant. This appendix records the three identities on which the reconstruction rests, together with the relation between the two determinant families and the explicit low-order members.

The adjacent-layer identity. To apply [20, Lemma 2.1] to (22), take its variables $x_1 = x$, $x_2 = -t$, its parameter $a = \sigma$, and its row and column spectral parameters $p_i + b - a$ and $q_j + a - b$. Introduce its auxiliary variable s by multiplying the nonconstant entry in (22) by

$$\exp\left[s\left(\frac{1}{p_i + b - a - \sigma} + \frac{1}{q_j + a - b + \sigma}\right)\right].$$

The constant entry c_{ij} is independent of all variables. The row and column factors in the notation of that lemma can be chosen as

$$\begin{aligned} \varphi_i^{(\ell)} &= a_i (p_i + b - a - \sigma)^\ell e^{(p_i+b-a)x_1 + (p_i+b-a)^2 x_2 + s/(p_i+b-a-\sigma)}, \\ \psi_j^{(\ell)} &= b_j \left(-\frac{1}{q_j + a - b + \sigma}\right)^\ell e^{(q_j+a-b)x_1 - (q_j+a-b)^2 x_2 + s/(q_j+a-b+\sigma)}. \end{aligned} \tag{A1}$$

Their product divided by $p_i + q_j$ is the nonconstant Gram entry. Differentiation verifies the relations (15)–(16) of [20]; for example, the layer increment multiplies that entry by $-(p_i + b - a - \sigma)/(q_j + a - b + \sigma)$, and its difference from one is $-(p_i + q_j)/(q_j + a - b + \sigma)$. The auxiliary derivative multiplies it by $(p_i + q_j)/[(p_i + b - a - \sigma)(q_j + a - b + \sigma)]$. These are precisely the layer and auxiliary-variable relations in the lemma. Its first bilinear identity, restricted to $s = 0$ and $x_2 = -t$, gives (23).

Continuous-layer correspondence. For the continuous family (6), choose $c_{ij} = \delta_{ij}$, $\alpha_i = -(p_i - a)\rho_i$, and $\beta_j = (q_j + a)^{-1}$ in (21). At $b = a$, the limit with $kh = y$ is then $\tau^{(0)}_{1-n}$. Thus the layers $n = 0, 1$ converge to $f = \tau_1^{(0)}$ and $g = \tau_0^{(0)}$, respectively. For general b, the same correspondence holds after replacing x by $x - 2(b - a)t$. This accounts for both the reciprocal layer convention and the amplitude factors.

Rank-one identity. Let $M(\theta) = M_0 + \theta\, r\, c^{\mathsf{T}}$ with M_0 invertible and r, c column vectors. Factoring M_0 out of the rows of $M(\theta)$ gives

$$\det M(\theta) = \det M_0 \left(1 + \theta\, c^{\mathsf{T}} M_0^{-1} r\right), \qquad \partial_\theta \log \det M(\theta) = c^{\mathsf{T}} M^{-1} r. \tag{A2}$$

The second form follows from the first together with the Sherman–Morrison formula for M^{-1}, and it is the only identity needed to differentiate a determinant in a parameter that enters linearly.

Shift identity. The two families (29) and (34) differ only in their phases. The endpoint exponent is $(p_i + q_j)x + (q_j^2 - p_i^2)t$, and the centered exponent includes the additional term $-h(p_i + q_j)t$. The endpoint family gives the centered one through the simultaneous substitutions

$$\tau_n^{\mathrm{C}}(k; h) = \tau_n^{\mathrm{E}}(k; h)\Big|_{\substack{a \mapsto a + h/2 \\ p_i \mapsto p_i + h/2,\ q_j \mapsto q_j - h/2}}, \tag{A3}$$

These substitutions preserve $p_i - a$, $q_j + a$ and $p_i + q_j$, so the layer factors and Cauchy minors agree. Their effect on the endpoint phase gives precisely the centered phase in (34), retaining the corresponding positivity conditions.

Logarithmic fields. With $f_k = \tau_0(k)$ and $g_k = \tau_1(k)$ the definitions (40) and (48) become ratios of neighbouring determinants,

$$u_k = 2\partial_x \log \frac{\tau_0(k)}{\tau_1(k)}, \qquad v_k = \frac{2}{h}\partial_x \log \frac{\tau_0(k)\tau_1(k)}{\tau_0(k-1)\tau_1(k-1)}, \tag{A4}$$

and analogously for u_k, v_k through (48). Positivity of sufficiently smooth τ functions is a sufficient condition for smooth real physical-field reconstruction. Proposition 2.3 provides this condition for the parameter class (36). On any compact region in (x,t), with a fixed finite set of lattice indices, these determinants have a positive lower bound and their derivatives are bounded. The logarithmic derivatives defining the fields and their required derivatives are therefore bounded there. More generally, nonvanishing real τ functions permit the same local derivative formulas with $\log|\tau|$; field smoothness alone does not require a positive choice of τ.

Low-order members. For $N = 1$ and $c_{11} = 1$, the common family takes the form

$$\begin{aligned}\tau_n^{(b)}(k) = 1 + \frac{\alpha\beta}{p+q}\left(-\frac{q+a}{p-a}\right)^n \\ \times \left(\frac{1+h/(p-a)}{1-h/(q+a)}\right)^k \exp[(p+q)(x-2(b-a)t) + (q^2-p^2)t].\end{aligned} \tag{A5}$$

Here $b = a$ for Endpoint-SD and $b = a + h/2$ for Centered-SD. Substitution into (A4) reproduces the one-soliton field at its node and edge locations. The two-soliton case in Section 5 uses the corresponding 2×2 determinant in (21). The family with $c_{ij} = \delta_{ij}$ is the Gram reduction of the Casoratian determinants of the hierarchy [15].

Appendix B. Field transformations and reconstruction conditions

Endpoint field equation. Use the logarithmic ratios (43) and identities (44)–(45). For the second field equation, subtract the average of the first equation in (44) and its backward shift from the second. The required identities are

$$\begin{aligned} q_k - \frac{r_k + r_{k-1}}{2} &= \frac{h}{2}\Delta_h \log(f_k g_k), \\ q_{k,x}^2 - \frac{u_k^2 + u_{k-1}^2}{8} &= \frac{h}{8}(u_k + u_{k-1})v_k + \frac{h^2}{16}\left[v_k^2 - (\Delta_h u_k)^2\right]. \end{aligned} \tag{B1}$$

Using (45) to combine the second derivatives, and multiplying the difference by $4/h$, we obtain

$$\begin{aligned} 0 = 2\partial_t \Delta_h \log(f_k g_k) + (\Delta_h u_k)_x + (\frac{u_k + u_{k-1}}{2} + 2a - 2h)v_k \\ + \frac{h}{4}v_k^2 - 2(u_k + u_{k-1}) - \frac{h}{4}(\Delta_h u_k)^2. \end{aligned} \tag{B2}$$

Differentiating in x and using $v_k = 2\partial_x \Delta_h \log(f_k g_k)$ proves (42).

Centered field reconstruction.

The numerical state of Centered-SD is the pair (u_k, ω_k), whereas the physical comparison is made in the fields (u, v). This appendix records the two transformations between them, the conditions under which they are equivalent, and the lower trace required by the computation.

Field relation and recurrence. The definitions (48) imply, for every edge,

$$v_k - \Delta_h u_k = \frac{2}{h}\left(\omega_k + \omega_{k-1}\right), \qquad \omega_k = \partial_x \log \frac{g_k}{g_{k-1}}, \tag{B3}$$

where the second expression shows that ω_k is the lattice difference of the logarithmic potential and therefore carries a factor h relative to $v - u_y$. Rearranging (B3) gives the reconstruction (50),

$$\omega_k = \frac{h}{2}\left(v_k - \Delta_h u_k\right) - \omega_{k-1}, \qquad k = 1, \ldots, N. \tag{B4}$$

The recurrence is linear and lower triangular, so it determines $(\omega_1, \ldots, \omega_N)$ uniquely from the physical fields and the single lower value ω_0. Its solution is

$$\omega_k = (-1)^k \omega_0 + \frac{h}{2} \sum_{j=1}^{k} (-1)^{k-j} \left(v_j - \Delta_h u_j \right), \tag{B5}$$

in which the coefficients have modulus one. An error in the lower trace alone propagates with alternating sign and unchanged magnitude, while errors in the edge differences can accumulate through the sum. The reconstruction is an exact affine relation and requires no branch selection.

Conditions on the lower trace. For the τ functions of Theorem 2.1 the compatible trace is fixed by the lattice, namely $\omega_0 = \partial_x \log(g_0^{\mathrm{C}} / g_{-1}^{\mathrm{C}})$. In a computation the data are prescribed in the physical fields, so the trace is taken instead from the continuous reference, $\omega_0 = h(v^* - u_y^*)/4$ evaluated at the lower node. The two agree to $O(h^2)$ on the smooth class of Section 3.4, and the resulting discrepancy enters (B5) with alternating sign only, that is without growth along the layer index. During the evolution the lower value is advanced by the same equation as the interior, so the trace is not frozen.

Consistency of the initial data. Equation (B4) reproduces v_k on every edge by construction. Once the lower trace is supplied, (u, ω) represents the prescribed physical fields (u, v). Changing the lower trace by $\delta\omega_0$ and repeating the recurrence adds $\delta\omega_k = (-1)^k \delta\omega_0$. Its adjacent values sum to zero, so the reconstructed initial v_k remains unchanged. This alternating auxiliary component can affect the subsequent evolution; the lower trace is therefore specified together with the physical initial data. All computations in Section 5 use the construction above.

Appendix C. Operator compatibility and continuum expansions

We record the operator expansion used in the proof of Proposition 3.3 and the remaining field-reconstruction calculations. The continuum expansions used in Section 3.4 follow below.

Operator expansion. The definitions of r_k, q_k and the x derivatives of (44) give

$$\begin{aligned} 2\partial_x^2 \log f_k - 2\partial_x^2 \log f_{k-1} &= 2(q_{k,xx} - r_{k-1,xx}), \\ r_{k-1,xt} &= r_{k-1,xxx} - 2(b + r_{k-1,x}) r_{k-1,xx} - 2\partial_x^3 \log f_{k-1}, \\ q_{k,xt} &= q_{k,xxx} - 2(b + q_{k,x} - h) q_{k,xx} - 2\partial_x^3 \log f_k. \end{aligned} \tag{C1}$$

Applying the product rule at each site yields

$$\begin{aligned} & \big((L_t + [L, M])\psi\big)_k + 2q_{k,xx}(L\psi)_k \\ = \; & \left[2\partial_x^2 \log f_{k-1} - 2\partial_x^2 \log f_k + 2(q_{k,xx} - r_{k-1,xx})\right] \partial_x \psi_{k-1} \\ & + \Big[r_{k-1,xt} - r_{k-1,xxx} + 2b r_{k-1,xx} + 2\partial_x^3 \log f_{k-1} \\ & \qquad + 2r_{k-1,x}(\partial_x^2 \log f_{k-1} - \partial_x^2 \log f_k) + 2r_{k-1,x} q_{k,xx} \Big] \psi_{k-1} \\ & + \left[-q_{k,xt} + q_{k,xxx} - 2(b + q_{k,x} - h) q_{k,xx} - 2\partial_x^3 \log f_k \right] \psi_k. \end{aligned} \tag{C2}$$

The first and last coefficients vanish by (C1). Substitution of the first relation in (C1) into the middle coefficient reduces it to the second relation. This proves (62) componentwise.

Recovery of the nonlinear systems. For Endpoint-SD, differentiate $u_k = 2r_{k,x}$ in time and use (44) to obtain the first relation in (64). The last relation of (45) gives its second relation. Their lattice difference gives (41). For the second field, substitute (44) into

$$h v_{k,t} = 4q_{k,xt} - u_{k,t} - u_{k-1,t}. \tag{C3}$$

Using $q_{k,x} = (u_k + u_{k-1} + h v_k)/4$ reduces the result to (42).

For Centered-SD the definitions give

$$q_{k,x} - r_{k,x} = \omega_k, \qquad u_k = 2r_{k,x} + \omega_k. \tag{C4}$$

Subtracting the x derivatives of the two relations (44) yields (53). Adding (53) to twice the x derivative of the first relation (44) yields (65). The reconstruction gives $\Delta_h(2\partial_x^2 \log f_k) = \partial_x(v_k + \Delta_h u_k)/2$, so the lattice

difference of (65) gives the first equation in (49). For the second, differentiate $v_k = \Delta_h u_k + 2(\omega_k + \omega_{k-1})/h$ in time and substitute (53) and (65). The second derivatives combine through $-v_{k,xx} + 2(\omega_{k,xx} + \omega_{k-1,xx})/h = -\partial_x^2 \Delta_h u_k$, giving the second equation in (49).

Midpoint expansions. For smooth q sampled at the nodes and edges of the lattice,

$$\Delta_h q_k = q_y + \frac{h^2}{24} q_{yyy} + O(h^4), \qquad \frac{q_k + q_{k-1}}{2} = q + \frac{h^2}{8} q_{yy} + O(h^4), \tag{C5}$$

evaluated at the edge midpoint. Substituting (C5) into the first Endpoint-SD equation gives the first relation in (66), since both the difference and the average are second-order accurate there. The second equation carries an additional term from the flux. Expanding its bracket at the midpoint gives

$$(u + 2a)v - 4u + u_{xy} + \frac{h}{4}\left(v^2 - u_y^2 - 8v\right) + O(h^2), \tag{C6}$$

and applying ∂_x and solving for v_t gives the correction $-\frac{h}{4}\partial_x(v^2 - u_y^2 - 8v)$ in (66).

Centered reconstruction. The relation $v_k = \Delta_h u_k + \frac{W_k + W_{k-1}}{2}$ with $W_k = 4\omega_k/h$ expands as

$$v = u_y + W + h^2 \left(\frac{u_{yyy}}{24} + \frac{W_{yy}}{8} \right) + O(h^4), \tag{C7}$$

so on the smooth family with $W = v - u_y + O(h^2)$ the reconstructed second field is second-order accurate. The extra flux of the first equation is $\omega^2/2 - h\omega = h^2(W^2/32 - W/4)$, whose lattice difference and subsequent x derivative are $O(h^2)$, which gives the first relation in (67). Differentiating (C7) in time and using $W_t = W_{xx} - \partial_x[(u + 2a)W - 4u]$ from (53) gives the second relation in (67).

Appendix D. Numerical implementation and supplementary experiments

This appendix collects the conventions of the computations, the closures at the transverse ends, and the control runs quoted in Sections 5.3 and 5.4.

x derivatives. The fixed-grid spatial and time-integrator comparisons use periodic Fourier operators on $[-L, L)$. For mode m, their multipliers are $\mathrm{i}\pi m/L$ for $\widetilde{\partial}_x$ and $-(\pi m/L)^2$ for $\widetilde{\partial}_{xx}$. The first-derivative multiplier is set to zero at the Nyquist frequency, the second retains its value there, and products are evaluated pointwise. No filtering, dealiasing or artificial viscosity is applied in any run.

Closures. Endpoint-SD advances (u_k, v_k) with the prescribed lower value of u_t and a backward chain, so no upper datum is required. FD uses the same lower data and, at the upper end, the rule $u_{N+1} = u^*_{N+1} + 3e_N - 3e_{N-1} + e_{N-2}$ with $e_k = u_k - u^*_k$, together with its differentiated form when $\dot{u}$ is needed. Centered-SD advances (u_k, ω_k), needs no upper ghost, and advances the lower ω trace by the same equation as the interior.

Implicit step. Crank–Nicolson is solved by an Euler predictor followed by fixed-point iteration on the trapezoidal residual. The implementation advances (u, W), where $W = 4\omega/h$. In the notation of the field equations, the trapezoidal residual satisfies $\max\{\|r_u\|_\infty, (4/h)\|r_\omega\|_\infty\} < 2 \times 10^{-14}$ at termination. A control calculation with a stopping level of 2×10^{-15} changes the terminal error norms of Section 5.3 by at most 8×10^{-10}.

Moving mesh. The longitudinal derivatives use the explicit mapped three-point formulas (83), with $x_i = \xi_i + s_i$ and the discrete Jacobian defined there. The node motion is coupled to the field equations through $\dot{\boldsymbol{z}} = F(t, \boldsymbol{z}) + \mathcal{V}\widetilde{\partial}_x \boldsymbol{z}$. The uniform-mesh results of Table 5 use the same realization with $s_i = 0$, $J_i = 1$ and $\mathcal{V} = 0$.

The control runs underlying Sections 5.3 and 5.4 are summarized below. They use the same Centered-SD state, grid and reference as the corresponding main runs, and a single quantity is changed in each row.

The Euler half-step controls move the errors toward the RK4 values, supporting the cancellation and reinforcement interpretation in Section 5.3. The tighter implicit solve checks sensitivity to the stopping tolerance. The frozen-mesh controls show that most of the short-time gain in Table 5 is already obtained from the initial equidistribution.

Table 7: Control runs for Section 5. The change column states the only difference from the corresponding main run.

(a) Terminal field errors

Run	Change	E_u	E_v
One soliton, Euler	main, $\Delta t = 10^{-4}$	1.664×10^{-4}	8.026×10^{-5}
	half step, $\Delta t = 5 \times 10^{-5}$	1.671×10^{-4}	7.880×10^{-5}
Two solitons, Euler	main, $\Delta t = 10^{-4}$	1.420×10^{-4}	1.090×10^{-4}
	half step, $\Delta t = 5 \times 10^{-5}$	1.560×10^{-4}	1.048×10^{-4}
One-soliton mesh	adaptive, nodes moving	5.954×10^{-4}	1.157×10^{-3}

(b) Additional control results

Run	Change	Observed change
All, CN	tight solve, 2×10^{-15}	terminal error norms change by at most 8×10^{-10}
One-soliton mesh	equidistributed, frozen	reductions of 12.8% and 16.3% relative to the uniform mesh

Resolving the RK4 and Crank–Nicolson differences

Table 8 lists the saved terminal norms and their signed difference, C–N minus RK4; a positive difference favors RK4. The extra digits distinguish entries that coincide after rounding in Table 4.

Table 8: Terminal RK4 and Crank–Nicolson error norms at T = 0.01, with h = Δx = 0.1 and Δt = 10^{-4}. The last column compares the two saved main runs.

Case	Field	RK4	Crank–Nicolson	C–N - RK4	Smaller error
One-soliton	u	$1.67723337656 \times 10^{-4}$	$1.67723193356 \times 10^{-4}$	-1.44300×10^{-10}	C–N
	v	$7.73375836247 \times 10^{-5}$	$7.73376020056 \times 10^{-5}$	1.83809×10^{-11}	RK4
Two-soliton	u	$1.79163370095 \times 10^{-4}$	$1.80474053431 \times 10^{-4}$	1.31068×10^{-6}	RK4
	v	$1.00906108654 \times 10^{-4}$	$1.00948746633 \times 10^{-4}$	4.26380×10^{-8}	RK4
Gaussian	u	$2.89993671090 \times 10^{-8}$	$2.89992077157 \times 10^{-8}$	-1.59393×10^{-13}	C–N
	v	$6.96696389468 \times 10^{-9}$	$6.96879887130 \times 10^{-9}$	1.83498×10^{-12}	RK4
Sinusoidal	u	$2.00293429689 \times 10^{-8}$	$2.00290237694 \times 10^{-8}$	-3.19200×10^{-13}	C–N
	v	$4.70597516511 \times 10^{-9}$	$4.70499461613 \times 10^{-9}$	-9.80549×10^{-13}	C–N

The soliton rankings in Table 8 persist under RK4 step halving and a tighter C–N solve. For the Gaussian and sinusoidal tests, differences are only about 10^{-13}–10^{-12}, and RK4 step halving reverses some of the closest rankings. These entries describe the computed norms at the stated step size.

Appendix E. Non-soliton reference solutions

We construct the two non-soliton benchmarks by choosing a transverse parameter in a heat-equation representation of the DLW system. The construction uses a Laplace transformation of the associated Boiti–Leon–Pempinelli system, as studied in [16]. The parameter choice below produces the initial profiles of Section 5.1 and gives their subsequent continuous evolution.

Linear evolutions. Write $X = x - 4t$, $s(y) = A\sin[\pi(y+2)/4]/2$, and denote the prescribed longitudinal profile by F_0. Let C be its mean and choose $J_0(X) = CX + J_{\mathrm{per}}(X)$, with $J_0' = F_0$ and zero-mean periodic

J_{per}. Here $C = 0$ for the cosine and $C = \sqrt{\pi}/32$ for the periodic Gaussian. Treating s as an independent parameter, solve

$$\begin{aligned} F_t &= -F_{XX}, \qquad F(X,0) = F_0(X), \\ H_t &= H_{XX}, \qquad H(X,s,0) = -se^{-sJ_0(X)}. \end{aligned} \tag{E1}$$

The chosen profiles give explicit backward-heat evolutions,

$$\begin{aligned} F_{\mathrm{S}}(X,t) &= e^{(\pi/12)^2 t}\cos(\pi X/12), \\ F_{\mathrm{G}}(X,t) &= \frac{3}{\sqrt{9-t}}\sum_{j\in\mathbb{Z}}\exp\left[-\frac{(X+192j)^2}{4(9-t)}\right], \quad t<9. \end{aligned} \tag{E2}$$

The forward-heat solution H is obtained from the Fourier series of $e^{-sJ_{\mathrm{per}}}$, retaining the factor e^{-sCX}. Define ψ by

$$\psi_t = FH_X - F_X H, \qquad \psi(X,s,0) = e^{-sJ_0(X)}. \tag{E3}$$

Equation (E1) gives $\partial_t(FH) = \partial_X(FH_X - F_X H)$. Since $\psi_X = FH$ at $t = 0$, this identity holds throughout the evolution and fixes the spatial derivative needed in the reconstruction.

Physical fields and initial matching. Define the denominator

$$D = \psi H_s - H\psi_s.$$

In the following formulas, s is treated as an independent parameter when differentiating. In particular, u_s differentiates the first expression before substituting $s = s(y)$. The transformed fields are

$$\begin{aligned} u(x,y,t) &= -2\left(\frac{D_X}{D} - \frac{\psi_X}{\psi}\right), \\ v(x,y,t) &= 4 + 4s_y\frac{FD}{\psi^2} - s_y u_s, \qquad X = x - 4t, \quad s = s(y). \end{aligned} \tag{E4}$$

Substitution using (E1)–(E3) gives both continuous DLW equations (1), with $a = 2$ and $\lambda = -2$. The final expressions divide only by ψ and D, so zeros of the cosine profile or of s_y leave the reconstructed fields regular.

Initially, $H = -s\psi$ and $D = -\psi^2$. Consequently,

$$u_0 = 2sF_0, \qquad v_0 = 4 - 6s_y F_0 = 4 - 3u_{0y}. \tag{E5}$$

This calculation explains the joint choice of the two initial fields. The coefficient three is generated by the transformed family. It does not impose a constraint on the subsequent evolution: the DLW equations give $\partial_t(v - 4 + 3u_y)|_{t=0} = 8u_{0xxy}$, which is generally nonzero. At $y = \pm 2$, $s = 0$ gives $u = 0$, while $s_y = 0$ at $y = 0$ gives $v = 4$. The quasi-periodic factors in H and ψ cancel in (E4), yielding periodic physical fields.

Regularity and reference evaluation. For the amplitudes $A = 0.05$ and $A = 0.03$, we have $|s| \le 0.025$. Fourier coefficient bounds for (E1), combined with Cauchy estimates on the parameter disk $|s| \le 0.05$, control H, ψ and their parameter derivatives. Comparing them with $\psi_0 = e^{-sJ_0}$ and $D_0 = -e^{-2sJ_0}$ gives the uniform bounds

$$\psi > 0.6, \qquad |D| > 0.4, \qquad |x| \le 96, \quad |y| \le 2, \quad 0 \le t \le 0.01. \tag{E6}$$

For example, the coefficient estimates $\|F\|_{\mathcal{A}_2} < 2$, $\|J_{\mathrm{per}}\|_{\mathcal{A}_2} < 11$, and $0 \le C < 0.056$, where $\|q\|_{\mathcal{A}_2} = \sum_n (1 + |\xi_n|)^2 |q_n|$, give $|J_0| < 16.432$ on $|X| \le 97$. Integrating (E3) bounds the change in ψ; the forward-heat bounds control the change in H. Cauchy estimates for their s derivatives then bound the change in D, giving (E6). These lower bounds justify the smooth reconstruction on the full test region.

We evaluate (E4) through Fourier series and analytic parameter derivatives. Each heat mode has an explicit time multiplier, so the reference evaluation introduces no time-stepping error. The stored references use a degree-32 exponential expansion and 96 positive Gaussian Fourier modes; the sinusoidal input has one mode. Independent interval evaluation, including analytical tails and floating-point conversion, bounds the error in every stored initial field and time derivative and every stored final field at $T = 0.01$ by 10^{-13}. This bound applies at the native sample coordinates used in the error tables.

Acknowledgements

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Declarations

Competing interests

The authors declare that they have no competing financial interests. Additionally, there are no personal relationships or affiliations that could have influenced the work presented in this manuscript.

Author contributions

Four authors contributed equally to writing the paper.

Data availability

No datasets were generated or utilized in this study.

Conflict of interest

The authors declare that they have no conflict of interest.

Ethics approval and consent to participate

There are no human or animal participants in this article and informed consent is not applicable.

Consent for publication

The authors confirm the work described has not been published before.

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